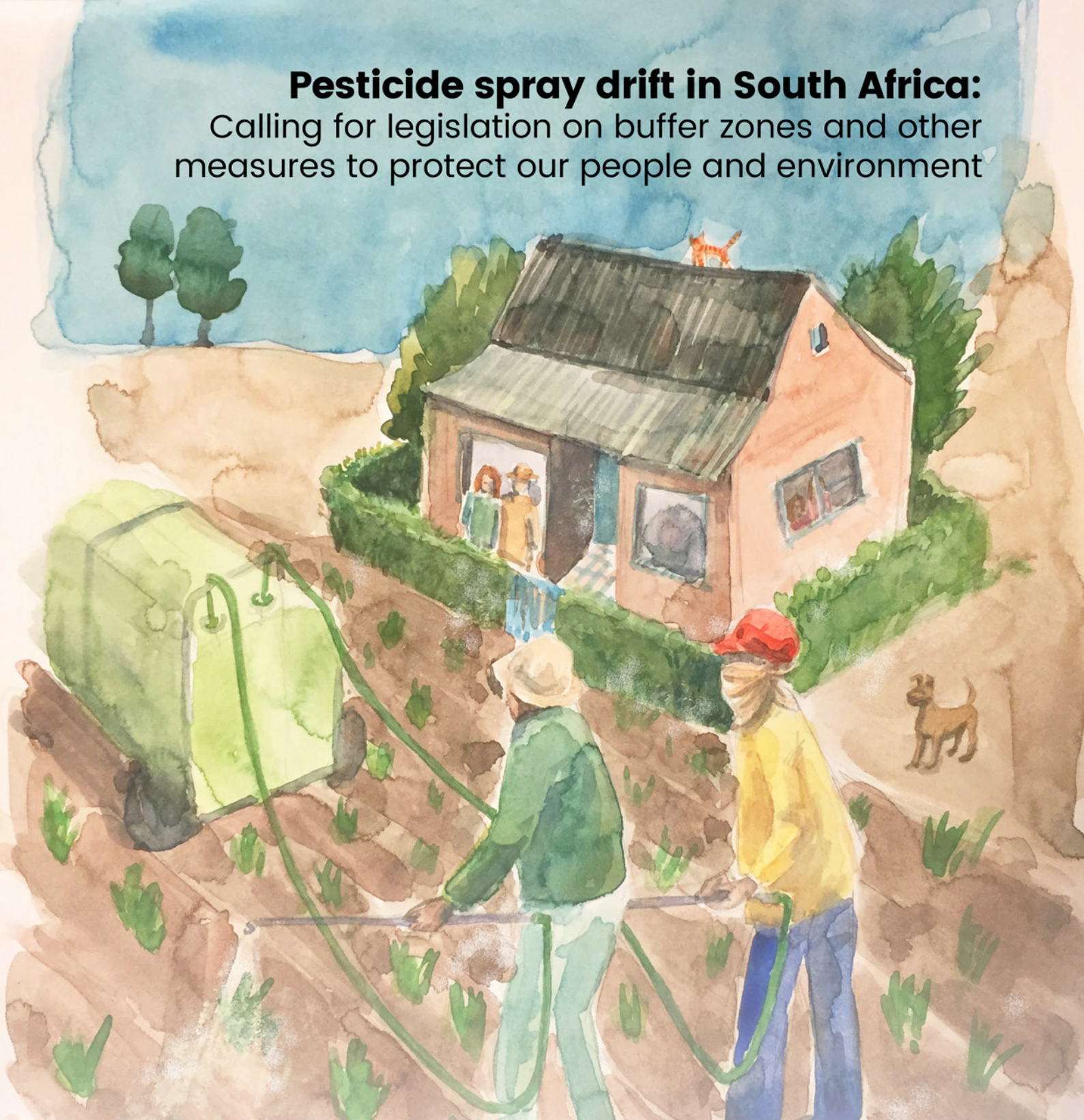


Pesticide spray drift in South Africa: Calling for legislation on buffer zones and other measures to protect our people and environment



Sasha Mentz-Lagrange & James Dabrowski



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Acronyms & initialisms

AC-WG	Agrochemical Working Group
ACB	African Centre for Biodiversity
AFSA	Alliance for Food Sovereignty in Africa
AGDISP	AGricultural DISPersal (model)
ANSES	Food and Environment Safety Agency (France)
APVMA	Australian Pesticides and Veterinary Medicines Authority
AR	Application Rate
CapSA	Child-health Agriculture Pesticide Study in South Africa
CMR	Carcinogenic, Mutagenic, Reprotoxic
DAFF	Department of Agriculture, Forestry and Fisheries
DALRRD	Department of Agriculture, Land Reform and Rural Development
DWAF	Department of Water Affairs and Forestry
DWS	Department of Water & Sanitation
EDC	Endocrine-disrupting Chemical
EDP	Endocrine-disrupting Property
EIS	Ecological Importance and Sensitivity
ELU	Existing Lawful Use
EPA	Environmental Protection Agency (USA)
GMO	Genetically Modified Organisms
GPfZ	GMO & Poison-free Zones
HCS	Hazardous Chemical Substance
IARC	International Agency for Research on Cancer
IDP	Integrated Development Plan
IPM	Integrated Pest Management
MoE	Margin of Exposure
MRLs	Maximum Residue Levels
MSDS	Material Safety Data Sheets
NOAEL	No Observed Adverse Effect Level
NOEL	No Observed Effect Level
NWA	National Water Act
OECD	Organisation for Economic Co-operation and Development
PAIA	Promotion of Access to Public Information Application
PAN	Pesticide Action Network
PCO	Pest Control Officer
PES	Present Ecological State
POP	Persistent Organic Pollutant
PPE	Personal Protective Equipment
PPP	Plant Protection Product
RAL	Regulatory Acceptable Level
SAOSO	South African Organic Sector Organisation
SAPS	South African Police Services
SDRAT	Spray Drift Risk Assessment Tool
TATIB	The Air that I Breathe
UNEP	United Nations Environment Programme
WHO	World Health Organisation
WUL	Water Use License

Summary

The health and environmental issues associated with pollution caused by agrochemical farming in South Africa go largely undocumented and are mostly disregarded by regulatory authorities.

The extensive use of agrochemical remedies on fruit orchards, vineyards and crop lands, which are cultivated extensively in South Africa, and the associated pesticide spray drift these lead to, is a serious yet under-reported issue. Rural communities are experiencing numerous and severe cases of agrochemical spray drift that affect the environment (accumulations of residues in soil, waterways and underground water that go largely unmonitored), severely affect human health, not only of the farm workers applying the remedies but also communities living adjacent to the sprayed fields. These practices also compromise the livelihood of farmers running agroecological organic operations (either by direct drift or by general contamination of the environment where farming occurs).

Spray drift is an externality associated with chemical farming for which no one is being held accountable. In South Africa, there is a dire lack of research into the causal linkages between this chronic exposure and illness, including non-communicable diseases and reproductive disorders. But there is abundant scientific evidence of such correlation in the international literature (London 2019).

South Africa's legislation on pesticides is regulated by Act 36 of 1947 on agricultural remedies. The 2010 Pest Management Act sought to address the limitations of Act 36 of 1947, with specific consideration given to human health and the environment. However, these provisions have to date failed to materialise in a tangible manner. Overall, the framework regulating pesticide usage in South Africa proves weak and insufficient as it narrows their regulation to spraying protocols and product specification, with still limited consideration given to occupational health, the health of adjacent rural communities, long-term environmental impacts and the need to sanction abuses and compensate victims.

The pesticide industry is highly influential and pressurizes the regulator for the industry's interests to prevail. At the crux of this matter is the fact that South Africa's agricultural system is locked in an agrochemical construct (IPES Food 2016), and that government, the industry and therefore farmers push the flawed notion that food cannot be grown without chemicals.

This paper documents three cases of pesticide drift contamination linked to the cultivation of table grapes, strawberries and blueberries in the Western Cape, and how these affect rural communities. These stories are a small drop in the ocean of spray drift issues plaguing the country. They all give a sense of the disempowerment and hopelessness felt by these local communities, who are virtually left with no recourse.

Given the projected growth of the berry sector in South Africa - a 95% growth in blueberries is for instance projected by 2023, with the mainstay of this growth expected to materialise in the Western Cape (Meintjes 2019) - it can be expected that the issue will only grow more dire in the coming years.

In January 2020, legal correspondence was addressed to the Presidency and several ministries, calling for an overhaul of the agrochemical legislation in South Africa, which it described as 'outdated, fragmented and ineffective'. This paper supports this call and brings further evidence of why this is needed, with a specific focus on the issues related to the spraying of pesticides. The lack of inter-governmental cooperation around the management of pesticides is a critical issue that needs to be addressed as a matter of priority. It is high time the South African Government revised and implemented the 2010 Pest Management Policy to protect the constitutional rights to clean air, clean water and clean food.

This legislation should be revised to set South Africa firmly on the path of pesticide reduction, starting by giving serious priority to non-chemical alternatives. Further measures the Government needs to take specifically to curb this spraying drift issue include: minimizing or prohibiting pesticide use in specific areas, mandating buffer zones to protect non-target rural communities, and safeguard zones where surface and groundwater is used for the abstraction of drinking water, establishing robust surveillance and control systems, putting in place a country-wide ban on aerial spraying, solid provisions for advance warning (of spraying), requiring the use of drift-reducing equipment and setting up sanction and compensation mechanisms so that social and environmental justice can prevail. In addition, as encouraging farmers to transition to softer alternatives is so critical to move towards pesticide reduction in the country, a “transition fund” should be established to support these farmers.

This paper shows how other countries have gone about mandating compulsory buffer zones to preserve people’s health and the environment, especially water bodies. These approaches are not ideal, with buffer distances often being judged as insufficient by civil society, but they are a start. The applicability of the Spray Drift Risk Assessment Tool (SDRAT) to determine the ideal dimension for buffer zones in South Africa is described. The paper also discusses the manner in which the current South African legislation, notably the Pest Management Act, the National Water Act and the Air Quality Act could serve as leverage points to start legislating on buffer zones.

“ The legislation should be revised to **set South Africa firmly on the path of pesticide reduction,** starting by giving serious priority to non-chemical alternatives. ”

Introduction

There are in excess of 3500 pesticide products approved for use in South Africa. Among these feature possible neurotoxic and endocrine-disrupting chemicals (EDCs), including active ingredients bifenthrin, chlorpyrifos, cypermethrin and mancozeb (Dalvie et al 2003; Dabrowski 2015), which are banned in Europe.

In the front line and most exposed to the deleterious effects of pesticides are farm workers. There are countless stories of farm workers working on commercial fruit or vegetable farms who are seen applying pesticides with limited basic protective gear (masks and gloves) or sometimes without any form of Personal Protective Equipment (PPE), as shown in this paper¹.

Next in line are human settlements and neighbouring farmland which are subjected to the drift of pesticide spraying, commonly referred to as 'spray drift', which can be defined as "the movement of spray droplets of a pesticide outside of the application site during, or shortly after, application" (APVMA 2017:3). In South Africa, people affected by spray drift have little if no recourse to having their constitutional rights to clean air and clean water protected. Agrochemical spray drift is a big issue, with off-target communities potentially suffering serious health consequences, which include non-communicable diseases and reproductive disorders.

In South Africa, the legal and regulatory framework regulating pesticide usage is weak, with limited consideration given to occupational health and long-term environmental impacts. This constrained regulatory environment, characterised by government departments being unsure of their mandates, means that abusive practices often go uncurbed. The pervasiveness of agrochemical practices in South Africa means that farmers wishing to grow crops agroecologically and organically and who are adjacent to sprayed field, have to sacrifice their own crops to buffer the rest of their production (Ferreira, personal communication, 2019). When this is not possible, their crops suffer contamination by spray drift, which compromises organic certification.

There may be cases of amicable co-existence between chemical farmers and their neighbouring communities. Some farmers may resent being the cause of distress and feel helpless to escape from this chemical sump in which they find themselves drowning (IPES Food 2016), and would try limiting the impact of their practices on non-target areas. However even these benevolent measures can prove insufficient, as pesticide spray drift can travel far afield. But there are also many cases of highly charged and inimical relationships between chemical farms and their neighbours. Farmers would then, to evade the issue, claim that the chemical 'remedies' are applied in compliance with the law and that these are legally registered pesticides in South Africa (Women on Farms, an NGO, have called for the ban of 67 pesticides used in South Africa that are banned in Europe). Often people are told they should not have property close to a farm if they are going to complain about the pesticides. When they receive complaints, chemical farmers often call in the pesticide industry, which provides them with technical guidance on the chemicals to apply. The poison industry always makes sure to protect their own interests and reiterate the right to apply registered pesticides. When called in, the poison lobby often seeks to intimidate complainants.

This research paper highlights how the South African Government is failing to support individuals concerned about their health and that of their children by reining in abusive spraying practices thereby ensuring social justice prevails. The current situation in which chemical farmers can get away with spraying pesticides in strong winds, without taking wind direction into account, several times a week, and up to two metres from borders shared with neighbouring land, is tantamount to a state of lawlessness.

1 Although the issue of high exposure of farm workers to pesticides often banned in other countries is critical and needs addressing, it does not constitute the primary focus of this paper.

The three cases of spray drift described in the paper, are symptomatic, and serve to illustrate the nation-wide scale of the problem in South Africa. The first case study relates the story of a rural community, with specific focus on an individual whose life was shattered by the spraying practices of his neighbour, a table-grape farmer. The second is also located in the winelands, and describes the spray drift suffered by local residents and school children, linked to the spraying of strawberry fields. The third narrates the impacts suffered by a rural community adjacent to a blueberry farm on the Garden Route.

These case studies exemplify how easy it is for chemical farmers to get away with spraying practices that adversely impact their neighbouring environments. At the emission point, pesticide drift is difficult to control² and contaminates land, public spaces and people's homes in an invisible manner, thereby causing a range of health reactions and possible serious illnesses. These cases represent numerous others occurring throughout the country. At the time of writing, the authors were also doing research on the impact of spray drift related to large-scale fruit operations in the Eastern Cape and in Limpopo.

The only way to protect non-target areas is to put in place buffer zones, as many countries in Europe have done. The paper discusses how some European countries have gone about defining the dimension of legislated buffer zones (Netherlands, Slovenia), with specific focus on France, which has recently gone through a comprehensive exercise legislating buffer zones, which came into effect on 1 January 2020. There is however universal discontent about this law, with farmers complaining it reduces the surface of their farming land, and civil society organisations arguing that the imposed distances are insufficient. But it serves as a first step in a process that South Africa, given the numerous instances of pesticide drift, should be looking at addressing.

We describe SDRAT, designed by the Australian Pesticides and Veterinary Medicines Authority (APVMA) to calculate recommended buffer zone distances that would ensure that the Regulatory Acceptable Level (RAL) for a non-target receptor (e.g. human health) is not exceeded. This tool, which was tested in the South African context, can prove useful to inform the determination of buffer zones for the country. The current South African legislation, notably the Pest Management Act, the Water Act and the Air Quality Act can also serve as leverage points to start legislating for buffer zones.

This paper calls on the South African Government to start getting serious about reducing the pesticide load in our environment³ and giving priority to non-chemical alternatives. It advocates crucial amendments to the legislation governing the use of pesticides. Specific, feasible and immediate interventions needed to contain incremental spraying regimes that will go hand in hand with the projected growth in fruit production in South Africa are discussed.

2 There are of course ways to reduce the risk of pesticide drift, as described in Box 1 below.

3 It is important that the discourse centres on pesticide reduction as opposed to Integrated Pest Management (IPM), under the banner of which abusive use of pesticide goes unhindered. "Pesticide reduction can be quantified and analyzed statistically, whereas IPM cannot" (Ehler et al 2000:1).

1. South Africa's agrochemical regime

The ubiquity of pesticide in the South African environment

There are in excess of 3500 pesticide products approved for use in South Africa. This number is set to increase. According to the 2013 Global Chemical Outlook (United Nations Environment Programme 2013), South Africa is the largest consumer of pesticides in Africa, accounting for 2 per cent of global pesticide consumption. South Africa's pesticide expenditure rose close to 60 per cent over the period between the late 1990's and 2009 and was projected to rise another 55 per cent in the period 2009 to 2019. Pesticide-use maps generated in 2015 showed strong correlations between the national agricultural area under maize and the average annual use of atrazine⁴ per hectare of agricultural land. Maps indicate high usage of the pesticide across the country (Dabrowski 2015). According to analysts, the upward trend in glyphosate use in the country is likely to continue, increasing the glyphosate levels present in the environment and potentially increasing animal and human exposure to the herbicide (Benbrook 2016), unless a ban on glyphosate brings this growth to a halt (African Centre for Biodiversity 2019).

In South Africa, studies have linked pesticide exposure to acute poisoning (Bennet et al 2013), acetylcholine esterase inhibition as a result of aerial application of organophosphates (Dalvie and London 2006), possible occurrence of Guillain-Barré syndrome in a rural farming community (London et al 2004), birth defects (Heeren et al 2003) and endocrine disruption (Aneck-Hahn et al 2007; English et al 2012) in human communities. But pesticide contamination with toxic effects doesn't only affect human beings or animals; pesticides also penetrate deep into aquatic and terrestrial ecosystems. Many pesticides (e.g. DDT, dieldrin, Endosulfan) have the effect either to bio-concentrate in organisms (which often happens in aquatic environments) or to be biomagnified in food chains (i.e. the levels of concentration rises through the consecutive levels of a food chain).

This eco-toxicity issue is critical to the debate, as maximum residue levels (MRLs) do not take into account the cumulative impact of pesticides in our environment and food chain, and especially not the ubiquitous reliance on pesticides (resistance to degradation through metabolic activity, ultraviolet radiation and extreme temperatures) (Quin et al 2011).

These effects are concerning from an environmental and human-health point of view, particularly in the latter case, considering that many communities in South Africa use groundwater for drinking purposes or do not have access to, or reliable access to, treated water and often drink untreated water, which may be contaminated with pesticides (Dabrowski 2015).

South African farm workers highly exposed

There are countless stories of farm workers working on commercial fruit or vegetable farms who are seen applying pesticides with limited basic protective gear (masks and gloves) or sometimes without wearing any form of PPE, as evidenced in this paper. Additionally, workers and farm dwellers have little to no access to health care, which makes documenting the causal link between exposure and illness very challenging. A Human Rights Watch report decried the precarious position in which many workers and farm dwellers found themselves in the Western Cape. It described how some farm workers only received gloves, overalls and dust masks for protection. These offer hardly any protection against chemicals (Human Rights Watch 2011).

4 A systemic herbicide that is moderately toxic to most aquatic life, earthworms and honeybees and can leach into water.

The pesticide exposure of farm workers is a serious alienation of the rights of employees in the workplace. Section 24 of the Constitution of the Republic of South Africa states that “Every person has the right to an environment which is not harmful to their health and well-being”.⁵

The reality is that even if they do wear the most sophisticated PPE, this equipment still fails to protect their health, as was emphatically underlined in the Pesticide Politics in Africa Conference held in 2019, and which concluded with the Arusha Call for Action on Pesticides:

Even (when applied) by responsible and trained users, pesticides cannot be used safely, given the insufficient protection offered by personal protective equipment (PPE) under real conditions of use.
(The Arusha call for Action on Pesticides, 2019:1)

The Bordeaux University, which for years has researched and observed the use of the most sophisticated PPE in France found that, under real conditions of use (heat, unavoidable dermal (skin) contact with contaminated clothes, washing of PPE in the same washing machine as ordinary clothes, etc.), this equipment failed to adequately protect human health (Garrigou, personal communication, 2019).

5 South African Constitution, Act 108 of 1996.

2. Pesticide drift: a colossal yet under-documented and unregulated issue

While the primary source of pesticide contamination remains food (by consuming non-organic fruits and vegetables), exposure also happens through inhalation and dermal contact. The risks posed by pesticide exposure are not restricted to farmers and agricultural workers who are in direct contact with these substances. Pesticide drift, which is the drifting of spray and dust from pesticide applications, can be difficult to control and can easily spread into public areas and people's homes if preventative precautions are not put in place.

In South Africa, there is a rising trend of residential homes being built adjacent to agricultural areas, and of agricultural areas significantly expanding and encroaching into agricultural residential areas. As part of the field research conducted for this paper in the course of 2019 and 2020, two cases of massive commercial expansion into export fruit production on the Garden Route were researched, with one written up as a case study (see Case study 3).

Where homes, schools or other places frequented by people are located near fields where pesticides are sprayed, humans can also be exposed. Pesticides can also deposit indoors and then become a source of recurrent exposure. The route of exposure can be through pesticides deposited on surfaces, furniture, floor and food. Children can also be exposed in school playgrounds when running, falling, rolling on park lawns and on sports grounds. A dog rolling in grass close to a freshly sprayed field will expose anyone who touches the pesticides that were applied. Pesticides are invisible and the deleterious health effects associated with direct or indirect contact (chronic exposure) can take years before manifesting (Pesticide Action Network Europe n.d). The only pesticide effects that attract the attention of authorities are large-scale and acute or result in death and are therefore reported in the media.

A study conducted in the French region of Gironde revealed that 100% of the sites tested were contaminated with wine pesticides, and in each site carcinogenic, reprotoxic and endocrine-disrupting residues were detected, including prohibited molecules (diuron, metalaxyl, mepanipirim, flusilazole). These sampling sites were located between 5 metres and 500 metres from the vineyard plots and none of the inhabitants were vineyard workers. In the classroom of the primary school 15 pesticide residues were detected, including 10 reprotoxic, six carcinogenic and four endocrine disruptors. three molecules which had been banned were also found: mepanipirim, which has been banned since April 2017, metalaxyl, banned in 2005, and diuron which was banned in 2008. The study shows that the dwellings closest to the vineyard plots had the highest concentration of pesticides per gram of dust and that the sample from a dwelling located 50 metres from organically treated vines was less contaminated than another dwelling located 50 metres from vines treated with synthetic pesticides (HAPPI Survey 2018).

There are very few studies in South Africa documenting the adverse impact of pesticide drift on human health. The causal link between pesticide use and health outcomes was documented in a 2003 study which looked into the birth abnormalities of children of women exposed to agricultural chemicals during pregnancy in the Eastern Cape Province (Heeren et al 2003). A 2012 study showed pesticide affected the reproductive health of young boys (English et al 2012). More recently, Chetty-Mhlanga et al (2018) have been leading a study investigating the association between pesticide exposure on the reproductive health and neurobehavior of 1000 school-going children living in farmland and non-farm environments in three different agricultural areas in the Western Cape: Hex River Valley (table grapes), Grabouw (apple and stone fruits) and Piketberg (wheat and fruit). The studies also examine co-exposures from the use of electronic media, and maternal alcohol consumption. This study, called the Child-health Agriculture Pesticide Study in South Africa (CapSA) "will be the first longitudinal study investigating the reproductive health effects on children of agricultural pesticides in current use" (ibid. 857). At the time of writing, the results of the study were still being processed. Research findings will go a long way in alerting communities and government sectors to the need to rethink and better regulate pesticide use, to protect children's health.

3. What the South African law says

The framework regulating pesticide usage is weak, as it narrows pesticide applications to spraying protocols and product specification, with limited consideration given to occupational health, long-term environmental impacts and sanctioning abuses.⁶

Every agricultural chemical remedy (fertilizers and pesticides including herbicides, nematocides, insecticides, fungicides) that is used and sold in South Africa must be registered in terms of Act 36 of 1947. Act 36 states that when agrochemicals are used, the product label instructions and special warnings must be followed scrupulously.

In recognition that Act 36 is outdated legislation, the Act was amended to take into account the evolving context of pesticide usage in South Africa. Consequently, in 2011 the Pest Control Operator Regulation amendment⁷ brought in additional regulations to Act 36. It includes provision for precautions to be taken to prevent the drift or spread of agricultural remedies from a treated area. The amendment also specifies the following: “the pest control operator must ensure that agricultural remedies are used (including the use in the workplace) in accordance with approved label and with good application practice” (Article 9). Section II.11. of the 2011 amendment which deals with spray drift, states: “a pest control operator must ensure that all reasonable precautions are taken to prevent the drift or spread of agricultural remedies from a treated area” (DAFF 2011).

However, these provisions remain weak because the main ‘instrument’ that regulates the spraying of pesticides in the country is the requirement of Act 36 of 1947 to use pesticides in accordance with the labelling (which is based on the Registrar’s approval of the pesticide’s registration). This means that a complainant will need to prove that the chemical farmer is breaking the law and has used the product against the product’s label and thus in contravention of Act 36 of 1947.

There are numerous reasons why ‘agricultural remedies’ are often not sprayed in accordance with the product label (and thus Act 36 of 1947):

- considerable spray drift spreads to areas not under treatment (adjoining properties, dams, rivers, etc.)
- farm workers applying the application are not registered pesticide operators
- farm workers applying the application are often not wearing the correct personal protection equipment
- spraying occurs in adverse weather conditions (wind/temperature/humidity)
- inhabitants in the surrounding areas are not notified before the spraying commences, and signs are not erected post-spraying warning people not to enter the area that has been treated
- withholding periods (as per the product label) have been ignored as a result of which the maximum residue levels (MRL) are also not adhered to (Schirmacher, personal communication, 2020)

Any misuses should be investigation by the South African Police Services (SAPS) for criminal offence (London 2019). The evidence uncovered whilst researching this issue showed that this hardly ever happened.

In 2010, the Pesticide Management Policy (DAFF 2010) was passed, with the aim to “systematically revise Act 36”. The policy does bring improvement to Act 36 as it explicitly recognizes the unintended consequences of pesticide usage. It is worth noting that it mandates buffer zones for pesticides use. Within this act, there is mention of having to erect prescribed warning signs prior, during and after spraying, so as to warn people that a certain area has been sprayed so that they know to stay out of the area. As by law a municipality has to put signs up so that people don’t come into contact with freshly sprayed areas, and any lack of compliance can be brought to a court of law.

⁶ See Mentz et al. 2019 for a consolidated critique of the current legislation governing pesticides in South Africa.

⁷ Government Gazette, 18 February 2011. 14 No. 34020 No. 98 (..) Fertilizers, farm feeds, agricultural remedies and stock remedies Act, 1947 (Act No. 36 of 1947) Pest control operator regulation.

But this policy hasn't brought about all the changes it had set itself to affect. It only marginally improves the limited regulatory scope of Act 36. Furthermore, the draft policy paper received extensive commentary, which was dismissed and that, if incorporated into the draft law, would have significantly improved the management of pesticides and better protected people and the environment from non-target spraying (London, personal communication, 2019).

The major legislation on pesticides (Act 36 of 1947 and the 2010 Pesticide Management Law) is located in the Department of Agriculture, Land Reform and Rural Development (DALRRD) [formerly the Department of Agriculture, Forestry and Fisheries (DAFF)] but other departments are expected to play a leading role in surveillance of pesticide usage. The Department of Employment and Labour inspector is expected to enforce paragraph 9 of the Occupational Health and Safety Act (Republic of South Africa 1993) and enforce paragraph 10.3 of the Hazardous Chemical Substances (HCS) Regulations (Department of Labour 2003). This means that the main responsibility of ensuring that the use of pesticides does not pollute the atmosphere and injure third parties lies with the Department of Labour (London 2019).

However, labour inspectors will regard problems of bystander exposure to pesticides applied in agriculture to be a DALRRD matter and defer to law enforcement under Act 36 of 1947, which requires the SAPS to prosecute a farmer who applies a remedy contrary to the provisions of the label. For this to happen, an affidavit needs to be drawn up at the local police station, following which the DALRRD will be called to investigate. But they essentially investigate compliance with the Act in terms of protocols relating to spraying pesticides and not applications that go against the product label. If a breach to the Act is noted, then it is up to the public prosecutor to take action.

The 'London report' which served as scientific evidence to the 2020 legal request for agrochemical reform, comments on this as follows:

Besides the fact that the police, overburdened with high rates of crime in South Africa, will be unlikely to prioritise unlawful application of pesticides when having to investigate high rates of robbery and violent crime, the labels rarely provide sufficient detail to demonstrate intentional misapplication by a farmer. The DWAF (Department of Water Affairs and Forestry) inspectors will only investigate if evidence of contamination of water by pesticides is evident and local authorities require specific bylaws to empower environmental health practitioners. (2019:32)

Essentially, this means that it is only in instances of water contamination (e.g. a spill) of a water body that DWAF will intervene. And often identifying the source of pesticide contamination can prove challenging, especially in areas of intensive farming, where many farms could potentially contribute to water contamination.

One would expect that local municipalities should be in a position to intervene in problematic circumstances of spray drift but local authorities require specific bylaws to empower environmental health practitioners. Moreover, there might be political reasons why municipalities might be reluctant to intervene; in rural towns, commercial farmers are often affluent and influential on the local councils. They are key political entities to be reckoned with, and seen as a source of economic dynamisms in rural towns.

In the instance of Case study 1, the complainant described how the municipality kept on minimizing the claims brought about by the spray drift issue, whereas in Case study 2, the complainant suspected that the municipality's (he engaged with the environment officer and the planning department) claim that it couldn't do anything about the situation, had a lot to do with the political ties of the commercial farmer with the ruling party.

Given this current framework, we can only conclude that "rural communities effectively have no remedies for such exposure situations" (London 2019:32), as is further explained.

4. Struggles for social and environmental justice around pesticides in South Africa

Limited legal and institutional recourse of communities impacted by spray drift

Awareness about spray drift by the general public remains limited as it proves difficult for affected parties to speak out about this issue. People living close to sprayed areas feel they have no recourse and when they try to reach out to public entities for social justice, they often receive disappointing responses. Local municipalities and district municipalities respond that such issues fall outside their jurisdictions (although the issue of close proximity between commercial farming and residential living for instance points to disjointed planning). The institutional and legal void in which rural communities find themselves when trying to deal with this situation is framed as follows by London (2019):

In practice, rural communities have found that, when facing a situation of contamination resulting from drift, they are unable to find an enforcement authority sufficiently confident of their jurisdiction to take action against the pollution. This is partly the result of the personnel being unsure of the law and of their obligations under their laws, but also because of the fragmentation, overlap and regulatory gaps which exist in the current environment. (London 2019:31)

The Department of Agriculture's mandate is to ensure compliance with Act 36, which essentially entails ensuring the farmer keeps a spraying schedule, uses a registered pesticide applicator, and complies with the product label. But because farmers are loath to disclose the names of the products they spray, the farmer ends up being the sole jury of his compliance with the product label and the inspector can only conduct spot checks. Except for Case study 1, in none of the other cases researched for this paper did agricultural officials investigate an issue related to spray drift.

In the third case study described in this paper, affected parties reached out to various individuals in the local municipality, district municipality, to the Blue Scorpions, to the Department of Agriculture. But nothing proved the commercial blueberry farmer wrong and the expansion of his chemical-intensive activity in the Garden Route continues unabated.

This issue of the lack of recourse for social and environmental justice, which is the direct result of an inept regulatory framework, is a breach of Section 24 of the Constitution, which guarantees their 'right to a safe and healthy environment'. Pesticide drift needs to be understood from a legal perspective as undermining constitutional rights and this is how advocacy movements should put the issue to Government.

Dismissive and, at times, intimidating responses from chemical farmers, and the poison lobby

Affected communities are also disempowered by the responses they may get from the farmer. Chemical farmers can get defensive on the issue and often emphasize that they are farming in their own right, as the area is agricultural (implying prevailing interest or right of first mover) and that spraying was done in accordance with all regulations and all safety measures being taken. Even if the farmers genuinely comply with the regulations, the issue is that these are insufficient to protect the public, and that enforcement of irregularities hardly ever happens (we received no reports of such sanctions being applied).

Another reason why the issue of spray drift remains so out of the public eye, is because complainants are often subjected to threat and intimidation. In 2007, a spray drift issue made the headlines for a short while when a legal battle erupted between a wine farmer and one of his neighbours in Riebeek Kasteel (Case study 1). The complainant describes how his ongoing legal battle with the farmer led to life threats (Schumacher, personal communication, 2020). The blueberry case study almost had to be retracted from this paper as the people who had stepped in to share their story felt threatened by the farmer (he threatened to drive farm invasions onto their property).

One of the authors of the paper also had experience with spray drift and attempted to follow the letter of the law to prevent spray drift from affecting her land. The details of the case are not included here, but, as illustration, the nature of such intimidation is captured in a sample of the letters received in response to her complaints:



Photo 1. Tractor spraying 2-4 D Amin on fields before planting oats for animal feed

In correspondence dated 25 September 2013, the farmer responded with the following to the letter addressed to him complaining about the spray drift:

Kindly note that should you make any allegations regarding environmental or health implications resulting from our farming practices, you will be required to substantiate your claims by providing scientific data and I will furthermore consider these allegations as defamatory and I would take all steps necessary to protect our good name and standing by whatever legal means necessary.

His letter was accompanied by a letter written by Gerhard Verdoorn, who is the spokesperson of the Griffon Poison Information Centre. The letter reads as follows (extracts):

The organic people tend to blame farmers for the virus and bacterial infections found in the air and that is unacceptable (...) The farmer must point out to the complainant that should (his farming operation) suffer any damage to crops, pest invasion, plagues or pathogens as a result of the organic farming because of a lack of control thereof, he will be held civilly responsible for any and all losses incurred.⁸

A formal complaint was lodged with the local police station and prompted an investigation by the agricultural services. After the case had been left in limbo for a few years, the public prosecutor determined there was insufficient proof to substantiate the case and no charges were pressed.

⁸ Original in Afrikaans: “Die organiese mense het die neiging om virus en bakterie-infeksies van die lugweg op die boere se skouers te plaas en dis onaanvaarbaar (...) die boer die klaer daarop moet wys dat as daar enige peste, plae, onkruid of patogene vanaf die “organiese” boerdery na sy boerdery oorspoel as gevolg van geen beheer daarvan nie, dat die boer hom siviëlregtelik daarvoor aanspreeklik sal hou.”

Past and current mobilization by civil society to challenge pesticide issues in South Africa

The Air That I Breathe (TATIB) Foundation⁹

TATIB was formed in 2009 following numerous cases and complaints of unlawful pesticide spraying resulting in well documented spray drift in the Winelands of the Western Cape. It is a grassroots foundation that was originally focused on the areas of Stellenbosch, Franschhoek, Paarl, Somerset West and surrounding areas, as a platform from which the residents, parents, students, ratepayers, farmers and farm workers could campaign against the harmful effects of exposure to potentially toxic and unlawful spray drift from the adjacent vineyards and orchards, into bordering dwellings/residential areas. The foundation became less active when one of its co-founder members, Jurgen Schirmacher, whose story is captured in Case study 1, had to leave the country because of threats to his life that he received, due to his exposing of unlawful spray drift, his High Court victories to prevent unlawful spraying and because of his opposition to the chemical industry. TATIB played a key role in having Edosulfan banned for all uses (agricultural and domestic) and in banning chlorpyrifos for domestic use. To this day, TATIB still receives requests for assistance from all over the country.

Women on Farms petition government

In the Western Cape, a group of women farm workers called Women on Farms delivered in 2019 a petition calling for a ban of 67 pesticides that the EU has already banned. To date, there has been no response from Government moving forward with implementing the moratorium (Environmental Justice Atlas 2020).

The African Centre for Biodiversity (ACB)¹⁰

The ACB has historically been involved in researching and objecting to the capture of the African continent's agricultural sector by industrial agriculture. This includes fighting the repeated applications for the release of GMO maize varieties that exhibit resistance to multiple herbicides. In August 2019, the ACB issued, jointly with the Alliance for Food Sovereignty in Africa (AFSA), a call to ban glyphosate in Africa, which was supported by many organisations across the continent (ACB 2019).

GMO & Poison-free Zones (GPFZ)

GPFZ, is a citizen-based initiative created in South Africa to serve as a catalyst to nurture (and reclaim) land not yet prey to chemical farming methods through identifying and mapping 'uncontaminated' land throughout the country, and documenting cases of spray drift (including pesticide-related health issues), and contamination of organic crops by incursions of Genetically Modified Organisms (GMOs) in people's fields. This association has the potential of becoming the primary channel through which parties affected by spray drift and GMO contamination might have their voice heard, and lobby government for change. It is a body that intends holding farmers, authorities and the agrochemical industry accountable in the country. It is hoped this can be done by making visible the agroecological management of properties by earmarking them clearly in signage reading "This is a GMO and poison-free property – please respect buffer zones"¹¹

9 See: <https://tatibfoundation.blogspot.com>

10 See: <https://www.acbio.org.za>

11 See: <https://gmopoisonfreezones.org.za>

Request for reform of SA's agrichemical policy framework

On the 24th October 2019, a private entity (the Real Thing - Organic Supplements Company) served a request for legislative reform of SA's agrichemical policy¹² to the President and Deputy, as well as to the Minister of Agriculture, Land Reform, and Rural Development, the Minister of Health, the Minister of Environment, Forestry & Fisheries, the Minister of Employment & Labour, the Minister of Trade & Industry, the Minister of Water & Sanitation and the Minister of Finance. This Chemical Remedies Submission on Law Reform was substantiated by: a Chemical Remedies report by Prof. Leslie London, called the 'London' report (London 2019), a list of banned agrichemicals and a table of endocrine disrupters. Several departments acknowledged receipt thereof but discussion came to a brutal halt with the Covid-19 pandemic. Civil society is currently re-mobilising around this initiative.

The AgroChemical Working Group to #Unpoison South Africa

In 2019, several organisations and NGOs came together to work on supporting the rise of agroecology in South Africa. In 2020, the platform consolidated its work and organised itself into several working groups, one of which being the policy working group. A sub-group emerged under the policy group, with the specific focus of reforming South Africa's agrochemical legislation. This working group has since mid-2020 actively engaged with numerous stakeholders with the aim of raising public awareness of the impacts of agrochemical usage in South Africa. The campaigning emerging under this work is called #UNPOISON (our food, our air, our soils, our water, etc.).

12 Available from: <https://therealthing.co.za/index.php/in-the-real-submission>

5. Case studies documenting spray drift in the Western Cape

South Africa is the leading pesticide user in Sub-Saharan Africa (Curchod et al 2019) and a large quantity of the pesticide used in the country is concentrated in the Western Cape, which harbours fertile agricultural lands and is the mainstay of the country's wine and fruit production.

Three case studies of communities suffering the effects of intense spray drift in the Western Cape are documented. The spray drift suffered is told from the vantage point of people living or spending time right next to the chemical farms being sprayed. Save for the first case study, which was extensively described in the media between 2007 and 2008, the specific geographical areas of case studies 2 and 3 have been withheld and all interviews were kept anonymous, so as to protect their identity. Photos showing the proximity of the drift are shared as well as aerial maps showing the proximity of off-target areas impacted by the drift.

The first case study is an intense, personal story. The other two case studies show how people have and are being affected, but in a far less acute manner. We thank Jurgen Schirmacher for accepting to share his story for this paper and for all his photos.

Case study 1: Intensive spraying in the heart of Riebeek Kasteel

Jurgen Schirmacher's¹³ house was located 15 metres from the table grape vineyard sprayed by farmer John Vlok. When Schirmacher purchased his house, the adjacent land was uncultivated and did not belong to Vlok – in other words there was no vineyard. Schirmacher took Vlok to the Cape High Court in September 2006 after he suffered personal injury from the spraying. His new-born son became critically ill from the spraying. Schirmacher related his story in the writing (edited for concision):



Photo 2. Tractor spraying grapes 1 metre from residential fence (12 October 2004)



Photo 3. Poison spray from tractor blowing over Kloof Street (16 February 2006)

13 Schirmacher became a strong activist, campaigning against spray drift in the winelands. He co-founded the TATIB Foundation – See: <https://tatibfoundation.blogspot.com>

“when my wife was pregnant with Kristofer, in the summer of 2003/4, the farmer started spraying his precious newly planted vineyards with pesticides. When he sprayed, by means of tractor-pulled centrifugal ‘mist blowers’, our entire garden and house would be engulfed in clouds of spray drift.” (he asked the farmer what was sprayed and the farmer dismissed him, saying that nothing he sprayed was toxic).

“At the age of just five weeks, Kristofer fell ill with respiratory symptoms. We took him to a specialist in Cape Town, who examined him and immediately had him admitted to Constantiaberg Mediclinic. (The infant) went into cardiac and respiratory arrest later the same day (and had to receive 45-minutes of cardiopulmonary resuscitation. The doctors managed to stabilize him, and he was placed in intensive care where he remained for just over a week, having several seizures that first night. (He went back to hospital several more times during the next two years.) Tests confirmed that my wife’s exposure to agricultural chemicals, in particular organophosphate neurotoxins, had had a negative impact on the unborn child. (...) Armed with this new evidence we approached the farmer demanding that he furnish us with details of what he had been spraying on the adjacent & surrounding vineyards. He naturally refused and so we were forced to get our attorney involved hoping to compel the farmer to give us the information we had requested. As it turned out, the mists that rolled into town were deadly. The farmer had been spraying toxic pesticides, many of which had been banned worldwide.”



Photo 4. His son in ICU at five weeks after suffering acute exposure to pesticides from the drift (3 July 2004)

Frightening footage of the tractor spraying in the heart of the town created a public outcry in the May 2007 Carte Blanche exposé ‘Poisoned Fruit’.¹⁴

There was an application in September 2006 to the Cape High Court for an interdict stopping Vlok from spraying unlawfully. As discussed above, the only way a legal recourse can be sought is through a civilian court. In this case, the Director of Public Prosecutions brought formal charges against Vlok for spraying. Vlok and Schirmacher came to a settlement agreement, but after Vlok broke their agreement, Schirmacher took him back to court in early 2007 and the settlement agreement was made an Order of the High Court.

14 The exposé is accessible here: <https://www.youtube.com/watch?v=XshxTEovIU>

Despite the 2007 High Court Order, the farmer continued to spray in contravention of the law, and the terms imposed on him by the court (which included: giving notification at least 24 hours in advance prior to any spraying, providing the names of the product and spraying according to the law, and preventing spray drift out of his vineyards and into our house, garden and the town), “and so we were forced to take him back to court several more times.”

Throughout this whole episode, Schirmacher battled to secure local authority intervention (London 2019), and this despite the existence of a bylaw in the West Coast Municipality¹⁵ which empower environmental health practitioners to investigate issues related to air quality. He explains: “When I initially started complaining about our exposure to spray drift, the government and municipality did bugger all. They tried to sweep it all under the mat. They called me a serial complainant. They said that the giftowwe (poisons) were not toxic and that they were being sprayed according to the law etc. They went on to say that I would have to prove that the giftowwe were harmful to human health and that my health and that of my family had been impacted negatively by our exposure to what we were sprayed with. In true bureaucratic form they would simply tie one up with red tape.”

To win their legal battle, Schirmacher’s legal team opted not to refer to the physical injuries that had been suffered; they knew “(the respondents) had lots of money to pay experts to say that what they were spraying would not pose any threat to human health, and that we had not been affected negatively with regards our health. (...) We simply stated, to the judge, that it was clear that there was spray drift (our photographic proof clearly showed this) and that as a result the product label had not been adhered to, and that Act 36 of 1947 had been broken and that the farmer had thus broken the law.” They showed that the pesticides that had been sprayed were classified as highly hazardous by the World Health Organisation. “We therefore made it very easy for the Judge to find in our favour.”

“(An) exposé (by several journalists in newspapers & magazines about the spray drift) angered the farmer so much that he then started victimizing us. We found ourselves on the receiving end of death threats, crank calls, rocks thrown onto the tin roof of our house and even had shotguns fired at our house, from the adjacent vineyards in 2007 & 2008. Nothing was done, by the local police to enforce our court orders, nor prevent our lives being further threatened. We were forced to apply for an urgent Supreme Court Interdict in 2008 , hoping that this would deter the farmer (...).”

Their lives between 2004 and 2008 is replete with extreme incidents attributable to the spray drift. In December 2007, he ended up in ICU “after going into cardiac and respiratory arrest after being caught in spray drift of an unknown chemical. I spent six days in hospital and 14 days in bed. It was later confirmed, in writing, that I had been caught in spray drift of a organophosphate neurotoxin.” His wife developed pesticide-induced arthritis and was hospitalised for a few weeks. He went to specialists and was told that they (the Schirmacher family) would not survive another season of toxic spraying.

“In mid-2008 I was alerted to the fact that a contract had been taken out on my life in order to teach me a lesson for embarrassing the farmer and tarnishing the image of the town.

15 West Coast District Municipality. Air-quality management by-laws. Province of the Western Cape: Provincial Gazette No 7170 of 6 September 2013. Available at: <https://opengazettes.org.za/gazettes/ZA-WC/2013.html>

A taped conversation was lodged with the Directorate of Public Prosecutions in Cape Town. We immediately put the house on the market and at the same time laid criminal charges against the farmer”.

After moving in 2009 to Somerset West and despite still receiving death threats, Schirmacher founded, in October 2009, an environmental NGO named The Air That I Breathe [TATIB] Foundation.

“Our last 10 months in the country were spent campaigning against unlawful spray drift and the use of obsolete and often banned agricultural pesticides. Through the Legal Resources Centre, Cape Town, we lodged submissions to Government and managed to get the law tightened and get Chlorpyrifos & Endosulphan restricted and/or banned for use in South Africa. The farmer reciprocated by using his influence and ‘political’ clout to make our lives a living hell, despite the fact that we had left the town and were by then living a 90-minute drive away.”

Schirmacher ended up emigrating to Australia in 2010 due to continued threats and intimidation. But the health consequences of this intensive exposure to agrochemical drift between 2004 and 2008 persisted.



Photo 5. Jurgen Schirmacher's burnt chest after wearing a T-shirt off a washing line in his garden that had been caught in spray drift. The doctor's certificate stated the cause as 'acute contact dermatitis secondary to poison exposure'.



Photo 6. Aerial spraying (20 January 2006)

“In 2012, Michael (my eldest son – born in October 2001) and I were diagnosed with a very rare form of skin cancer that the specialists told us, was as a result of the dermal burns we received in 2008 in South Africa.” These chemical burns were caused by their washing on the line getting caught in spray drift. When they wore the T-shirts, their perspiration re-activated the pesticide residue on them, causing burns to their bodies. The diagnosis by specialists at the time was ‘acute contact dermatitis secondary to poison exposure’.

“We had surgery in 2012 to have it removed and have now been warned, by our doctor, to not expose ourselves to any chemicals, like chlorinated swimming pools, pesticides, etc..” Later that year (2012), he underwent two brain Computerised Tomography (CT) scans and a brain Magnetic Resonance Imaging (MRI), and various tests, and was told he would be needing two major brain operations. He has managed to hold off the craniotomies by adopting a very healthy lifestyle, losing weight and walking daily.

His son Kristofer has a delayed puberty due to his pre-natal exposure to organophosphate poisons. These are known developmental and reproductive toxins.

Former resident of Riebeek Kaastel, David Bellamy, explains that the political climate in Riebeek Kasteel has since then changed a little, with farmers, “not behav(ing) with as much impunity as (they used to), but the problem of pesticide drift persists, even occurring today.” Every season, from Wellington to Riebeek Kasteel, chlorpyrifos¹⁶ is sprayed by planes on the wheat fields. In winter, on a prophylactic basis and there is overlap of plane and tractor sprays. “We get at least weekly exposure and the downwind plume is kilometres long so exposure is perpetual.” (Bellamy, personal communication, 2020).

Case study 2: Blueberry production for export: spraying its neighbours away

A community of farmers and residents on the Garden Route (Western Cape) saw their peaceful lives disrupted by the expansion of a commercial blueberry farm right on their doorstep in 2018.

The closest neighbour to the farm is a couple who describe themselves as ‘lifestyle’ farmers. Their house is located 40 metres away from the netting structures, with the outside workshop being located 10-30 metres away from where the spraying takes place. The pristine fynbos which used to cover their horizon was mowed down and by 2019, netting structures were erected all around the commercial operation. These netting structures, erected right up to their fence, shut down their view but not the cloud of pesticides blowing from the blueberry operation. They captured footage showing how the spray drift goes through the nets.

16 Chlorpyrifos is a pesticide known to be extremely harmful to young children. It is banned in the EU and there is intensive campaigning in the US to have it banned. It was banned for domestic use in South Africa but is still authorised for agricultural use.

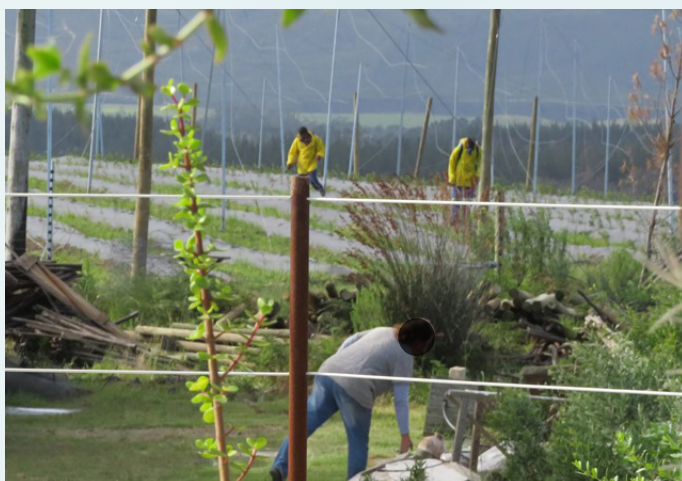


Photo 7. Shows the proximity of the blueberry farm to the resident's garden. The tall poles served as the supporting architecture of nets that were later erected two metres from the border. (Credit: anonymous, image captured 21 March 2018)

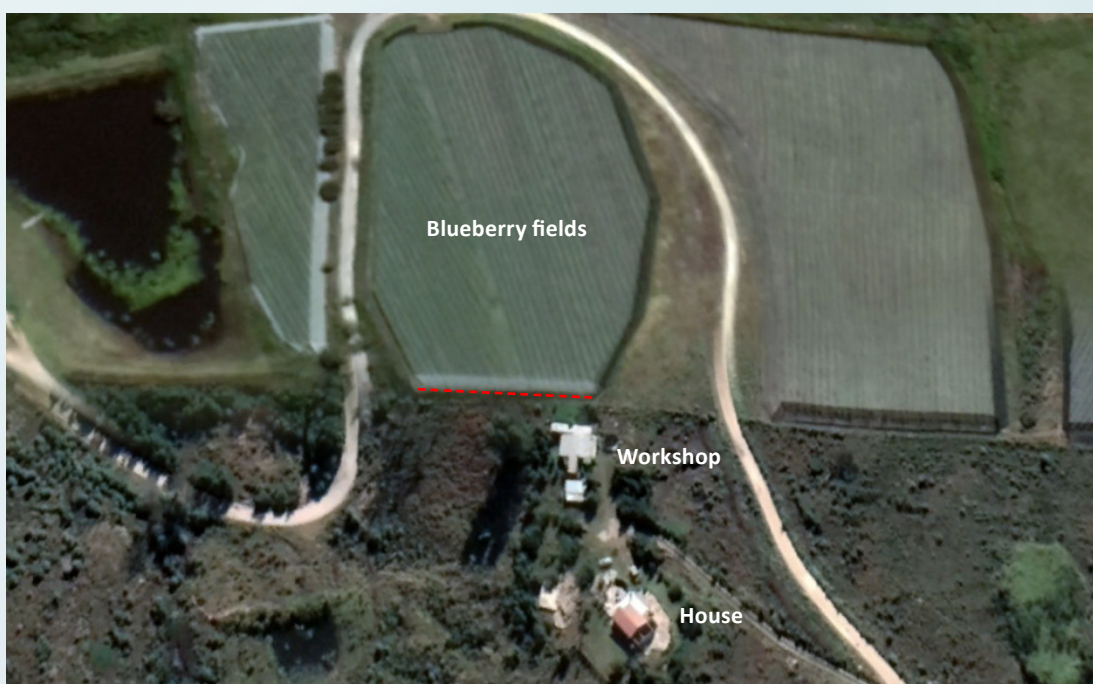


Photo 8. Aerial view of the dwelling, workshop and farming operation. (Source: Google Earth imagery dated 4 March 2020, accessed 28 July 2020) The other dwellings border the farming operation and access road.

Whenever the spraying took place, they reported suffering severe health issues, with symptoms ranging from watering eyes, to skin rashes and irritation, nose, throat and chest problems, difficulty breathing, numbness around the mouth and inside of the mouth, coughing, chest and joint pains, lack of concentration and headaches. They described the drift conditions on some days as so severe that it affects neighbours more than 400 metres away with similar symptoms. These clinical symptoms, which are typical of chronic exposure to pesticides, were compounded by major psychological anxiety imputable to the close proximity of farm workers spraying, and the dread of the next spraying episode. They described the impact on their lives as 'debilitating'. They would often have to leave their house when spraying happened. They ended up selling their property.

Having witnessed field workers and spraying staff working in the same area – most of the time without the full protection gear - they felt that contacting the Department of Agriculture inspectorate would play to their advantage. In response to the complaint, the inspector indicated before coming on-site "(but that) you must understand that we can only investigate within the framework of the Act. If the 'transgressor'/accused is spraying within the prescript of the label of the product, we can go no further." When the official came to conduct his inspection, he found the blueberry farmer was running an exemplary operation and decreed that the complaint was not receivable.

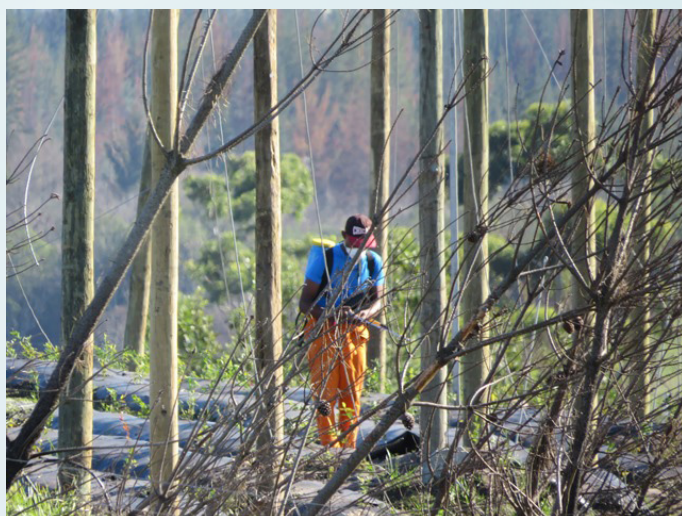


Photo 9. A farm worker spraying blueberry sprays with a rucksack. He is wearing only a mask, and no other form of PPE. (Credit: anonymous, image captured 12 March 2018)

This case study points to several key issues that are emblematic of the situation in the country. Firstly, the area was zoned as 'rural residential' by the local municipality, which led to lifestyle farmers cohabitating with commercial farmers. The extreme close proximity of the commercial farm operation to their border (1 metre from their fence, 30 metres from their house) points to a clear zoning issue; how can such cohabitation be possible and how can such encroachment into people's space be admissible? This begs the question of whose responsibility is it to protect residents from chemical exposure?



Photo 10. View from the complainants' garden once the netting structures were put up, one metre from the shared fence. (Credit: anonymous, image captured 3 November 2018)

The affected couple and their neighbours continually appealed to any administrative entities which in their view could play a role in changing this situation. They wrote to Water Affairs and the relevant Catchment Management Agency; both provided no response to their concerns about pesticide and fertiliser run-off into the town's main catchment area. The local and district municipalities were contacted in the hope they would address this issue, to no avail.

In early 2020 the couple sold their property, at a loss. The residents and farmers (including chemical farmers) who are still living there are feeling hopeless. Meanwhile the farm keeps on expanding its operations.



Photo 11. Area being cleared for the expansion of the farming operation on either side of the access road. (Credit: anonymous, image captured 21 October 2018)

Case study 3: Strawberry fields in Stellenbosch, Western Cape. “A desert being fumigated”

This is a large strawberry commercial operation. According to local residents, part or all the fruit production goes to Woolworths. The strawberries are grown under tunnels. The entire area planted to strawberries covers about 15 hectares. The strawberry farm bordered a wine estate which contains a pre-school and a primary school.



Photo 12. Aerial view of the strawberry farm, showing cottages and school. (Source: Google Earth imagery dated 2 April 2019, accessed 28 July 2020)

The spraying ‘season’ predominantly spans from June to November every year. Spraying is done with boom-sprayers mounted on tractors as well as with knapsack sprayers. Spraying occurs frequently – there are local reports of spraying taking place on a weekly basis from August to October, and neighbouring farms and the school do not receive any notice of the spraying, which has been reported to also occur in high winds.

There have been several reports of farm workers, school children, teachers and parents being affected by the spraying. In 2019, there was an unconfirmed report of a farm worker being rushed to hospital after visibly suffering from intense exposure. Photos of the spraying show workers spraying without any PPE. Spray drift in the majority of cases blows in the direction of a public road leading to the school.

The school is located about 800 metres away as the crow flies from the land where spraying occurs. However, the only access road to the school runs within 5-10 metres from the spraying area. When parents come and collect their children from school, the area where they walk is approximately two metres from where the workers spray (Anonymous, personal communication, 22 March 2020). This means potentially very proximate exposure to the spray drift for children, their parents and the school staff.



Photo 13. Tractor spraying the strawberries up to about eight metres from the main road. (Credit: anonymous, image captured 5 September 2019)

An individual testifying as part of a spray drift area report suffered congestion and nose bleeds on two occasions in 2019 (Anonymous. Pers. Com. 22 March 2020). Another mother of the children attending the school explained in correspondence that “My children are definitely experiencing the effects of the spraying. They are coughing (lungs) first symptom, their noses are sensitive, bleeding in my case and also (I) experience dry mouth, red eyes and a lack of energy. Brain-fog will also be one of the symptoms. The municipality will tell you that it the effect of pollen, but it is not true, I’ve worked with them in the past.” (correspondence, 27 August 2019).

One of the interviewees reported that there was little regard on the part of the farm operation for wind direction on what he described as “a desert being fumigated”. The interviewee used to live in a farm house alongside the strawberry fields with the farm house erected three metres from the shared fence, and the edge of the cultivated field being, in turn, 12 metres from the border. This equates to a 15-metre gap between spraying and the residential unit, despite which the interviewee reported smelling pesticides frequently, opting to leave his house when the spraying occurred and giving up any prospect of tending the garden organically given the extent of the drift (anonymous, personal communication, 22 June 2020). A tributary to the main river meandering through this farming area runs about 700 metres past the top mark when pesticides are applied, with pesticides potentially running off into this river.

Surrounding residents and the school personnel have limited knowledge as to what pesticides are being applied. When one of the parents asked to be informed about the pesticides used, as well as the farm spraying schedule, the farmer reiterated that spraying took place in full accordance with all regulations (a point that photo 13 below, showing farm workers spraying without wearing protective equipment, contradicts) and responded that they were “unable to furnish this as we do not spray on a fixed schedule but rather when we have exhausted all non-chemical alternatives and the pest has reached a threshold where a chemical intervention is required”, implying that non-chemical alternatives were used, but not naming these nor the agrochemicals used, as was requested. The farmer did state that on the day the children suffered the symptoms, an insecticide with the active ingredient Spinosad was used, which has low toxicity for people and other mammals. Other informants indicated that methyl bromide¹⁷ used to be applied (anonymous, personal communication, 22 July 2020), a product which was banned in 2015.



Photos 14 and 15. Farm workers spraying unidentified pesticides without wearing adequate protective equipment (i.e. no mask, no gloves). (Credit: anonymous, image captured 7 November 2019)

It can be assumed that fungicides and herbicides are being applied, as these are the most frequent treatments for strawberries. One of the affected parties initiated correspondence with Woolworths asking them about the pesticides they allowed their producers to use, which was evaded by Woolworths. At this stage the affected community is opting to engage constructively with the farm to obtain the information which they feel entitled to obtain (pesticides being used, spraying programme) as well as requesting the farmer to notify the adjacent community when he sprays.

¹⁷ The primary use of methyl bromide is as a fumigant in soil to control fungi, nematodes, and weeds; in space fumigation of food commodities (e.g. grains); and in storage facilities. Methyl bromide is highly toxic. Studies in humans indicate that the lung may be severely injured by the acute (short-term) inhalation of methyl bromide. Acute and chronic (long-term) inhalation of methyl bromide can lead to neurological effects in humans (EPA 2000).

Our case studies reveal that many instances of spraying occur within extremely close proximity to habitations, sometimes within one metre from gardens. In case studies 2 and 3, the owners of the farm live in cities and not on the farm, which indicates that they are oblivious and disconnected from the reality of the spraying, and unharmed by it.

The general population in rural and peri-urban areas is being chronically exposed to synthetic pesticides both outdoors and indoors. As these synthetic pesticides have harmful effects on health, it becomes fundamental, in the interest of public health, to reduce the risk of exposure of the population to these toxic molecules.

These case studies do not document the environmental implications of these recurrent spray drift areas. In case studies 2 and 3, important concerns about the impact of the potentially contaminated water runoff into nearby river tributaries were mentioned. As discussed above, people felt helpless about their situation, as they were cognisant of the presence of multiple polluters impacting these water bodies upstream. Systematic sampling of pesticide concentration in soils and water adjacent to sprayed fields and their associated environmental risks should therefore also be done as part of follow-up research.

The research has also identified a distinct trend of commercial fruit farming expanding for the export markets. Analysts indicate that the country's "berry production, led by a rise in blueberries, has tripled over the five-year period between 2013 and 2017, from only 450 hectares under production to more than 1700 hectares in 2017" (Meintjes 2019). It is currently predicted that plantings of blueberries will increase by 95% by 2023. Most of this growth (61 to 66%) will occur in the Cape (ibid.). If this is an exciting prospect for commercial exports, it also brings scary prospects for the rural communities where blueberry operations will start mushrooming in a context virtually devoid of regulations and enforcement of pesticide drift.

6. Working towards a solution

Reducing pesticide dependency and the impacts of pesticide loads in the environment and on non-target organisms and people, should form part of South Africa's integrated pesticide management strategy. This should be combined with giving priority to non-chemical alternatives, minimizing or prohibiting pesticide use in specific areas and establishing buffer zones to protect non-target rural communities and safeguard zones for surface and groundwater used for the abstraction of drinking water.

The current agricultural sector in South Africa is completely at odds with this internationally recognised imperative to promote non-chemical alternatives (although this intent is mentioned in the 2010 Pesticide Management Act), as the policies supporting emerging farmers are often accompanied by an expectation that these farmers will engage in high-input agriculture, including the use of pesticides. The support they receive in the form of government subsidies are genetically modified (GM) maize seeds, which to yield need to come with a package of expensive fertiliser and agrochemical inputs (African Centre for Biodiversity 2018).

It is also clear that South African commercial farmers are deeply locked into the chemical farming paradigm (IPES Food 2016), which doesn't mean they do not feel concerned about how these practices impact their neighbours. Conversations with farmers reveal that many are scared of untying this noose that has been tightened by years of insidious pushing by the agrochemical industry - which is also deeply enamoured with GMOs - through creating dependency and a debt trap.¹⁸ Indebtedness and crop failure feature as leading reasons why many farmers across the developing world are committing suicide or other forms of self-harm. In India, it is estimated that 16,000 farmers die by suicide each year (Nagaraj 2008). The latest figures on farmer suicides are dated back to 2016 and since then the Indian government has been withholding any publication of farmer suicide under various pretexts (First Post 2019). But this issue is also rife in industrial countries, where the agriculture sector is modernised. In France, an average of one farmer commits suicide on a daily basis. Debt is often the cause but also isolation by men in their fifties or sixties who often don't have successors to take over the farm (Terre Net 2019).

Among the key possible interventions that the public needs to request from government as part of their constitutional rights (Section 24) to limit spray drift impacts, include: the need for legislating adequacy-sized buffer zones, the need to move to banning aerial spraying, the need for solid provisions for advance warning, and the need to require the use of drift-reducing equipment. As this paper focuses on buffer zones, we only discuss the other legislative imperatives briefly. As encouraging farmers to transition to softer alternatives is so critical to move towards pesticide reduction in the country, setting up a "transition fund" backed by technical expertise to support these farmers will also be critical.

Ban aerial spraying

South Africa should ban aerial spraying, as it exposes bystanders and workers in neighbouring fields, as well as inhabitants living in the outlying areas. Aerial spraying also causes increased drift.

There are many documented cases of the illnesses caused by aerial applications of glyphosate in Latin America. In Argentina, a study looking at the concurrence of glyphosate exposure and reproductive disorders in an Argentine agricultural town found that spontaneous abortion and congenital abnormality rates were respectively three and two times higher than the national average. This was due to the very high environmental exposure to glyphosate (79 kilograms per person per year) (Avila-Vazquez et al 2018).

¹⁸ See Auerbach (2020) for the policy shift required by the South African agricultural sector to achieve sustainability.

Aerial spraying has been banned in Slovenia. In South Africa, the aerial spraying of certain pesticides (2-4 D Amin, Dicamba and dimethylamine salt) was banned in 1991 in KwaZulu-Natal only (DAFF 2017). The government should not permit aerial spraying near natural parks, water bodies or close to residential areas.

Make spraying hazard-warnings, prior and post-application, compulsory

South Africa could draw from many international best practices to improve adequate information sharing when pesticides are sprayed. An effective prior notification system has run successfully for years in New Zealand. Farmers send spray application information up to an hour before spraying to a central data collection team. Interested members of the public can ask to receive this information on their local area by phone or text message. Farmers in Britain must tell beekeepers directly when they intend spraying. In Bulgaria the authorities must tell beekeepers as soon as they have been told about aerial spraying by farmers.

Mandate buffer zones

The Australian Environmental Protection Agency (EPA) defined buffer zones as “a zone around a pesticide application target area that captures off-target pesticide movement so it does not have an impact on more sensitive areas such as waterbodies, native vegetation or neighbouring properties” (Australian EPA 2017).

This means that a buffer zone plays an active role in remediating the negative impact of pesticide drift and run-off. It is a zone that is effectively removed from agricultural production to establish a permanent crop, e.g. grass, bushes and/or trees. It is therefore very different from the concept of ‘safeguard zone’ (or no-spray zone), which is commonly used to refer to a spray-free and/or fertiliser-free zone, which is otherwise treated in the same way as the rest of the field (Boye et al 2012).

As explained, although there is provision for buffer zones in the 2010 Pesticide Management Act, the South Africa Government currently does not make it compulsory for a chemical farm to set up a buffer zone or safeguard zone to prevent drift on non-target areas. Therefore, the onus is on the non-target neighbour to make provisions for such buffer zones, which are also described as “an area of separation between conflicting land uses” (Government of Western Australia Department of Health. n.d.).

Asked about the practices adopted in their organic citrus orchard, a farmer in the Eastern Cape indicated that they kept a buffer zone of 15 metres to protect their organic fruit. The buffer is taken from their land, with their own citrus trees forming part of the buffer. These are further protected on the outside boundary by tall windbreakers. The buffer zones seem functional in their specific set up, with wind breakers playing a key role in neutralising most of the drift, with three rows of fruit trees, then capturing the remnant of the drift. In different contexts, such as those discussed in the case studies, where there is no high vegetation sheltering rural communities from the drift, much wider buffer zones would be required to contain drift.

Making legal provisions for adequate buffer zones and drift-reducing spraying equipment would not only help reduce surface and ground water contamination, but could also help fight biodiversity loss (Pan Europe n.d.).

Developing key recommendations suited to the South African context, would also play an important role in signalling an imperative to contain spray drift, as was done by the EPA in Australia (Box 1).

Key recommendations for spraying pesticides (Australian EPA 2017)

“To reduce the chance of pesticides being washed off-target and into waterways, you should not apply them when there is rain forecast or when soils are already wet. Rain can also reduce the effectiveness of an application. Drift will be reduced when temperatures are mild and humidity is higher. Try to spray when there are consistent light winds of 3–15kph blowing away from sensitive areas.

Avoid spraying in low humidity and with temperatures above 27°C as spray droplet size will decrease due to evaporation, increasing the likelihood of drift. More volatile chemicals may also vaporise and re-enter the atmosphere. These conditions can also lead to inversion layers and unstable atmospheric conditions, greatly increasing the chance of drift.

Avoid very calm conditions, in which drift will be unpredictable, or strong winds, which may carry spray a long way.” (2017:55)

Establish an “agroecological transition fund”

Ultimately the pesticide burden on people and the environment will only be reduced if farmers phase out the use of agro-chemicals and adopt alternative practices. Because agro-chemical farming practices are so deeply entrenched in South Africa, it proves very challenging for farmers to transition to non-chemical practices. There is limited technical expertise available to support farmers in embarking in large scale conversion to agroecological and organic farming. A lot of it comes from organic farmers who are extremely busy on their own farms. Farmers wishing to transition would be greatly supported by greater expertise being made available to them, as well as a dedicated fund.

7. Review of best practices on buffer zones

With the new EU Directive on the sustainable use of pesticides (EC 2009), the issue of safeguard zones and vegetated buffer strips has become mandatory in all EU countries. We draw on a few examples from the EU to illustrate how these buffer zones have come about, with a specific focus on the recent developments that took place in France to legislate buffer zones.

The Netherlands: buffer zones determined by the types of crops grown

Since 2000, a law in the Netherlands makes provision for calculating the dimension of buffer zones based on the types of crops grown by the chemical farmers and the type of implements used by farmers. The value of the land is also a determinant in this calculation. Growers can either opt for a small buffer zone (0,5 metre) and adjust their spraying equipment to the most efficient to avoid spray drift, or they can adopt for wider buffer zones, in which case the requirements relating to their spraying equipment become less demanding (1-2 metres) (Pesticide Action Network Europe n.d.).

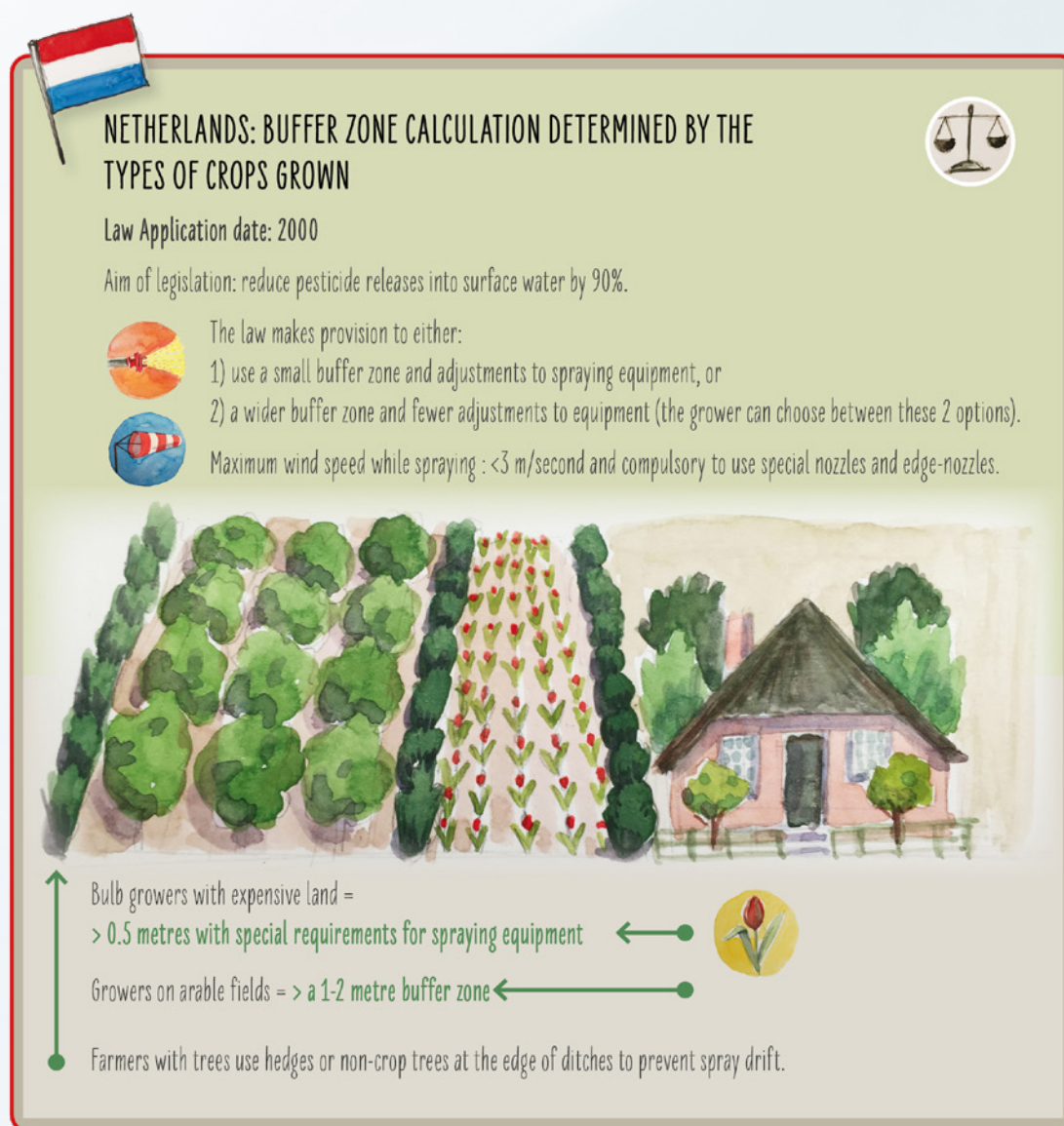


Figure 1: Netherlands: Buffer zone calculation determined by the types of crops grown

However, the distance still permitted under the provision of this law offers no protection to off-target sites and people. We saw, through the blueberry case study, what such an allowance for spraying within 0,5 to 2 metres from a neighbour's property resulted in.

Slovenia: buffer zones (with complete prohibition on the use of pesticides) defined by water protection regulations

More inspiring is the example of Slovenia, a country which has banned aerial spraying. The 2002 Water Act informs the dimension of buffer zones according to the categories of the water bodies.



Figure 2: Slovenia: Buffer zones defined by water protection regulations

The prescribed buffer zones are as follow: the buffer for coastal land and first-class surface water courses is 15 metres (complete prohibition on the use of plant protection products (PPP) in this zone) and the buffer for second class surface water courses is 5 metres. Furthermore, the use of pesticides and fertilisers is not allowed in first, second- or third-class underground water protection zones where a special regime has been established. Finally, wider buffer zones are defined in the Regulation as envisaged for specific plant protection products.

Heated debates around buffer zones in France

First steps to regulate the use of pesticides in the 2010s

In France, provisions to protect vulnerable people from pesticide drift date back to the decree of 27 June 2011. The decree prohibits the use of products hazardous to health in places usually frequented by children (school, crèche, day nursery and leisure centre, children's playground) and in the vicinity of establishments where elderly, sick or disabled people reside (hospitals and clinics, private health establishments, nursing homes, functional rehabilitation homes, establishments that accommodate or house elderly persons and establishments that accommodate disabled adults or persons suffering from serious pathologies).

Furthermore, a decree on the marketing and use of PPPs and their adjuvants promulgated on 4 May 2017, established a maximum wind limit of 19 km/h for pesticide application and the ban on any direct application of the product to elements of the river network. These include all watercourses, storm water retention ponds, gullies, gutters and catch basins.

Proposed amendments to the existing legislation

But these provisions fell short of protecting people within residential areas. At the behest of NGOs and concerned citizens, the government tabled an amendment on 23 June 2018 to introduce measures to protect people living near areas treated with PPPs as of 2020.

The new amendment provides for a framework for the use of pesticides near regularly inhabited residences and their adjoining areas (courtyards, gardens, etc.). It actually proposes to extend the provisions set up in the rural code to protect vulnerable persons (in schools, nurseries, hospitals, retirement homes, etc.) in residential areas.

The text lists a whole range of protective measures that need to be put in place. These must take into account the dangers of the products, the techniques and application materials used, and be adapted to the context (topographical, pedoclimatic, environmental and sanitary). Thus, the establishment of no spray zones near homes (as requested by some associations) is only one of the measures recommended. Farmers would be able to choose to comply with professional specifications validated by the state, or to respect treatment periods, dates or schedules where the use of spraying or powdering is prohibited. Another possibility available to them is to install physical protection infrastructure or the use of equipment to reduce drift.

The government is therefore banking on the introduction of proportionate measures and good practices by farmers, in order to promote the coexistence of activities in rural areas. Indeed, the establishment of buffer zones is decried by the National Farmers' Union, which considers that this would lead to a drastic reduction in the agricultural area that could be cultivated and treated. The establishment of these buffer zones will therefore be used by Prefects only as a last resort, the text states.

When debated in the Senate, many proposed amendments were submitted to regulate even more strictly the use of pesticides in the vicinity of housing. Some senators proposed the establishment of buffer zones of five metres minimum, near residential areas. In these zones, the use of carcinogenic, mutagenic, reprotoxic (CMR) or endocrine

disrupting products would be prohibited. In addition, the senators also proposed the implementation of clear signage to warn the public of treatment zones. Senators from the communist and ecologist group propose to extend restrictions on the application of pesticides in the vicinity of organic farming areas, in order to avoid any risk of contamination.

A bill tabled by MP Loïc Prudhomme (from the *France Insoumise* party) on 13 June 2018 is even more ambitious when it comes to protecting local residents. It provides for the establishment of 200-metre buffer zones around establishments receiving vulnerable persons, establishments receiving the public, and dwellings. In these zones, only low-risk treatment products would be permitted. This proposed law makes no provision for derogations such as anti-drift nets or adapted timetables from this ban, “as it appears that such arrangements do not sufficiently protect areas from pollution by these pesticides”.¹⁹

Provisions for minimal distance in the new law, applicable since 1 January 2020

On 20 December 2019, the State ruled on the minimum distances to be respected between areas treated with PPPs and residential areas:²⁰

- For the substances of greatest concern [any phytosanitary products labelled as carcinogenic, mutagenic, reprotoxic; those products that bear the following danger signs on their labels (in reference to EU regulation 1272/2008) : H300, H310, H330, H331, H334, H340, H350, H350i, H360, H360F, H360D, H360FD, H360Df, H370 and H372)]: **20 incompressible metres**²¹
- For other products: **10 metres** for arboriculture, viticulture, trees and shrubs, forest, small fruits and ornamental crops over 50 centimetres high, banana trees and hops;
- **5 metres** for other crops.

Those 10- and 5- metre thresholds can be lowered to 5 and 3 metres when farmers use techniques to reduce drift or exposure to drift by at least 66% compared to the normal conditions of spraying. In the case of viticulture, the buffer zones can be reduced from 10 metre to 3 metre when the reduction rate reaches 90%. This reduction does not apply to places housing vulnerable people (retirement homes, schools, etc.).

This law was not positively received on both sides of the spectrum. French farmers decried the Ministry’s announcement and some indicated that they would never follow the regulation unless duly compensated for any losses associated with the loss of land or losses incurred from pests. Environmentalists and NGOs decried the fact that despite being more protective than before, these provisions were still insufficient for those living near fields. Some associations demanded greater distances: 10 metres for all crops, at least 20 metres for viticulture and at least 50 metres for arboriculture.

Moreover, the obligation to inform the public about the nature of the products is still not made compulsory by these edicts, so local residents will have to “trust” the farmer who carries out the spraying. In view of these shortcomings, the associations *Agir pour l’Environnement and Générations Futures* indicated they would be taking legal action.

Several mayors passed legislation banning the use of pesticides in their town. But administrative courts struck down these local laws because banning pesticide use is not within the regulatory powers of local authorities.

19 See: [Proposition de loi n° 1069 protégeant la population des pesticides par l’instauration d’une zone tampon](#)

20 The proposed distances were based on the June 2019 proposal of the French Food and Environment Safety Agency (ANSES).

21 The same 20-metre buffer zone applies to products containing any active compound considered an endocrine disruptor.



FRANCE: BUFFER ZONE CALCULATION DETERMINED BY THE TYPES OF PESTICIDES USED, THE TYPES OF CROP GROWN AND THE SPRAYING EQUIPMENT USED.



Prohibition to apply any hazardous products in places frequented children (since 2011)



Maximum wind speed while spraying : <19km/h (since 2017)

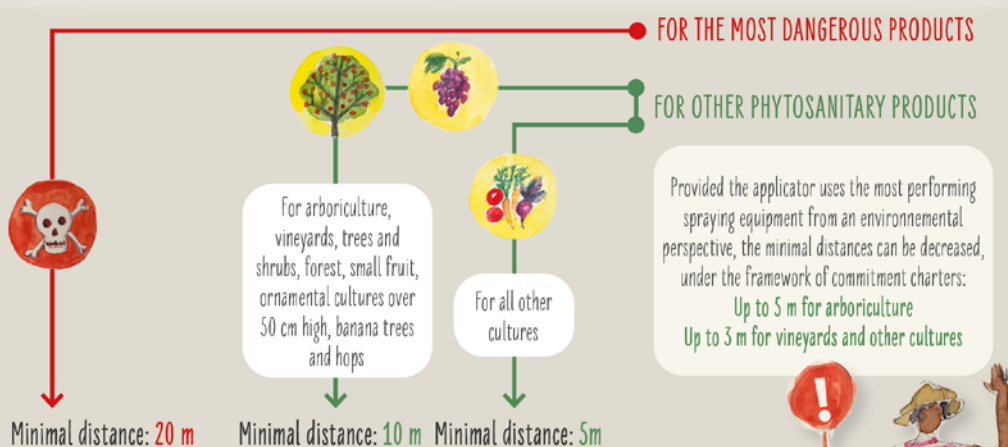


Ban on direct application to elements of river networks (since 2017)

MINIMAL DISTANCES

Between the treated areas and the residential areas

Law Application date: 1 January 2020



Note: these distances are deemed highly insufficient by civil society organisations, with some advocating for the establishment of 200-metre buffer zones around establishments frequented by vulnerable persons, public buildings and residential dwellings. In these zones, only low-risk treatment products would be authorized (Draft bill, tabled by France Insoumise).



Figure 3: France: Buffer zones calculation determined by the types of pesticides used, the types of crops grown and the spraying equipment used

Risk-based determination of buffer zones for protection of human health based on the Spray Drift Risk Assessment Tool (SDRAT)

Each pesticide has unique chemical properties that determine its inherent hazard and its interaction with living organisms (e.g. aquatic biota, humans, vegetation, livestock etc.). Each pesticide is therefore not equally hazardous to all living organisms. For example, an insecticide can be highly toxic to insects and at the same time much less toxic to mammals or aquatic life. Similarly, a fungicide or herbicide will have a strong effect on its target pest but might have little effect on human health. Some pesticides do not cause significant concern because they have very low toxicity to non-target organisms, may break down rapidly in the environment and/or do not accumulate in animals' bodies. By contrast, there are also pesticides that are highly toxic to humans and other organisms as well as to their target pests.

The risk a pesticide poses to a non-target organism can be understood by considering the pesticide's inherent hazard properties combined with potential exposure of the organism to the pesticide. Both 'hazard' and 'exposure' are important in measuring risk. Even the most toxic chemical known presents no risk at all if there is no chance of exposure. In the same sense, a pesticide only modestly toxic to humans could present unacceptably high levels of risk if exposure to it were high. In setting limits of exposure, it is important to quantify the toxicity of a particular chemical. For nearly all chemicals, living organisms have the ability to tolerate trace exposures without harm and ecotoxicological studies determine the concentrations at which chemicals will result in a statistically significant toxicological effect (e.g. LD50 or the dose which will result in 50% mortality in a specific test species) or cause no toxicological effect (e.g. NOEL or No Observed Effect Level).

The mode of action and toxicity of a chemical, the environment and the physiology of non-target organisms that might be exposed all interact in complex ways to result in outcomes potentially different for each combination. When all of these features are considered together, a threshold of risk for each pesticide can be determined for human health. Exposures above the threshold are deemed not acceptable and those below it can be considered negligible because large safety margins are built in to each threshold.

The Australian Pesticides and Veterinary Medicines Authority (APVMA) developed a Spray Drift Risk Assessment Tool (SDRAT). The SDRAT was designed to calculate recommended buffer zone distances that would ensure that the RAL for a non-target receptor (e.g. human health) is not exceeded (APVMA 2017). The RAL is the maximum amount of spray drift exposure which is not expected to cause undue harm to a non-target receptor (e.g. human health). RALs are established after a safety factor (also known as uncertainty factor, level of concern, margin of exposure, assessment factor, etc.) is applied to a relevant toxicity endpoint level which is determined through the assessment of information used for satisfying the statutory criteria (safety and trade) of the legislation.

For spray drift, the level of potential exposure is assessed from downwind spray drift deposition data. Deposition data details how much of a pesticide is deposited from spray drift at various points downwind from the application area. Many studies have been done to find out how spray droplets behave in the atmosphere when moving downwind. These aerial application scenarios rely on modelling software called AGricultural DISPersal (AGDISP) developed by the US Government. AGDISP is a 'true model' based on first principles which allows the use of different parameters to examine spray release height, wind speed, temperature, humidity and many other parameters including relevant features of different aircraft. Summaries of the input parameters are included with each scenario. The scenarios use the AT502 as the representative fixed wing aircraft. Each scenario displays downwind deposition data for ASAE droplet size categories each modelled for three different wind speeds. The AGDISP model has been successfully tested and validated in South Africa (Nsibande et al., 2015) and deposition data derived from SDRAT are therefore assumed to be acceptable for the purposes of this report. (See Box 2)

How the AGDISP model was tested in South Africa

Air dispersion software models such as AGDISP have been developed for predicting off-target movement of pesticides via spray drift and are frequently used in environmental and human health risk assessments. AGDISP was developed in the US and as such required validation against field monitoring data in order for it to be employed with confidence in South Africa. In the study by Nsibandé et al. (2015), off-target pesticide drift was monitored during ground application of a pesticide mixture to a sorghum field in Mpumalanga Province, South Africa. Atrazine was used as a drift tracer. High volume air sampling onto polyurethane foam was conducted at six downwind locations and at four heights at each sampling point. Additional data, including meteorological information, required to simulate the spray drift with the AGDISP air dispersion model was collected. The model accurately predicted measured air concentrations of atrazine at up to **400 metres** from the point of application. The results of this study showed that the model, although developed internationally, can successfully be used to predict spray drift in South Africa and can be reliably used to predict exposure to pesticides for the purposes of environmental and human health risk assessment studies in South Africa and other developing countries.

Downwind deposition curves (a deposition curve is a graphic plot of deposition data) for each of three standard droplet size spectra for a typical aerial fixed wing application are shown for three different wind speeds (Figure 4). In Figure 4, zero on the horizontal axis represents the downwind edge of the application area. The numbers along the horizontal axis represent the distance in metres downwind from the application area. The vertical axis of the graph is scaled as a fraction of applied field rate – for this example the value of 0.01 near the bottom of the graph means 1% of the pesticide rate applied to the field.

When all the information is assembled in this way, one can readily see how protective no-spray zones can be determined. The point where the downwind deposition curve intersects a risk threshold is the distance downwind from the application area that must be protected. This distance is the recommended no-spray zones (or setbacks), which are areas where spraying should not take place between the downwind edge of the application site and an identified sensitive area. For example, an RAL of 0.02 of the applied field rate would correspond to a no-spray zone of approximately 100 metres for pesticides applied at a coarse droplet size (Figure 4). This example illustrates why the use of smaller droplet spectra requires larger no-spray zones than larger droplet spectra because smaller droplets drift further.

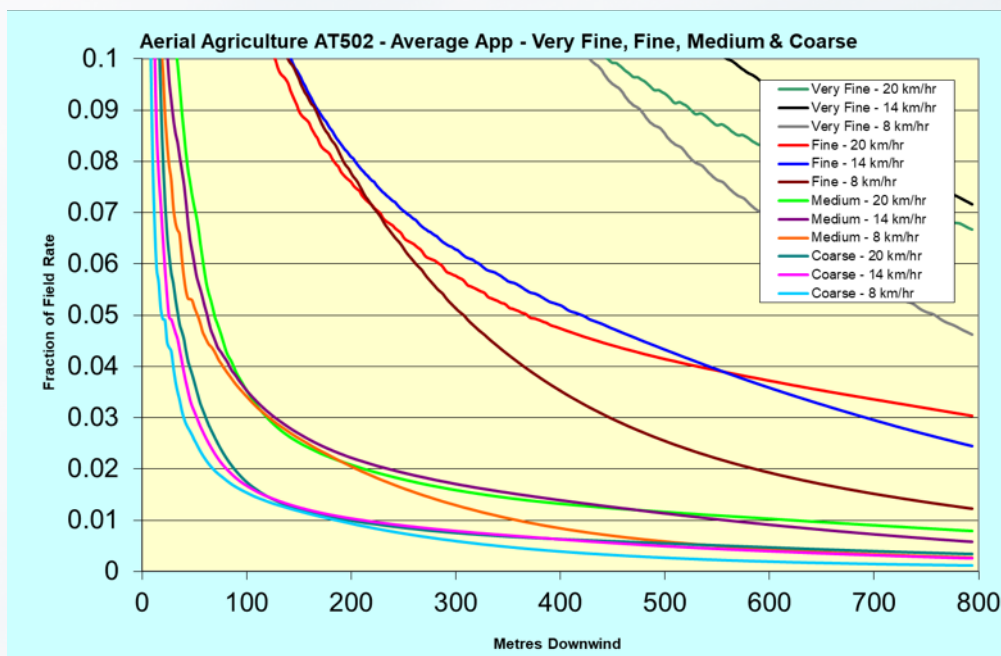


Figure 4: Drift deposition curves for different droplet sizes and wind speeds
Source: Nsibande et al., 2015

In the graph illustrated above, the medium and fine droplet spectrum would require a buffer of approximately 200 metres and up to 800 metres, respectively in order to remain within the risk threshold of 0.02.

Bystanders may be contacted by spray drift in two ways, either directly by the spray cloud, or, through repeat exposure to deposited residues on the ground. The method for estimating the RAL is based on that adopted by the US EPA (2014). Available tools used to predict spray drift are only capable of reliably determining deposition on the ground (i.e. a horizontal surface) which is not relevant for direct contact of bystanders with the spray cloud (i.e. a vertical surface). As it is not possible to estimate exposure reliably through direct contact by the spray cloud, pesticides should be applied so that direct contact with the spray cloud is avoided. Repeat exposure to deposited residues on the ground is akin to risk assessments that determine the risk of applications to turf so this risk assessment approach forms the basis for the determination of the RAL for bystanders. The highest concern for this type of exposure is that of an infant/toddler who may be exposed from repeat dermal and oral (hand-to-mouth actions) exposure whilst in a backyard or a garden over an extended period of time.

The formula used to determine risk after applications to turf is shown in Equation 1 (Annexure 2).

Models for estimating exposure and estimating risk to pesticides are commonly used in Europe, North America and Australia and form the basis of their regulatory framework. As highlighted above, the AGDISP model (and other pesticide fate and transport models), which forms the basis of the exposure calculations has been tested and validated in South Africa, although at a very limited scale and can therefore be useful tools for determining risk-based quantitative buffer zone widths in South Africa.

Australia's voluntary guidelines for buffer zones

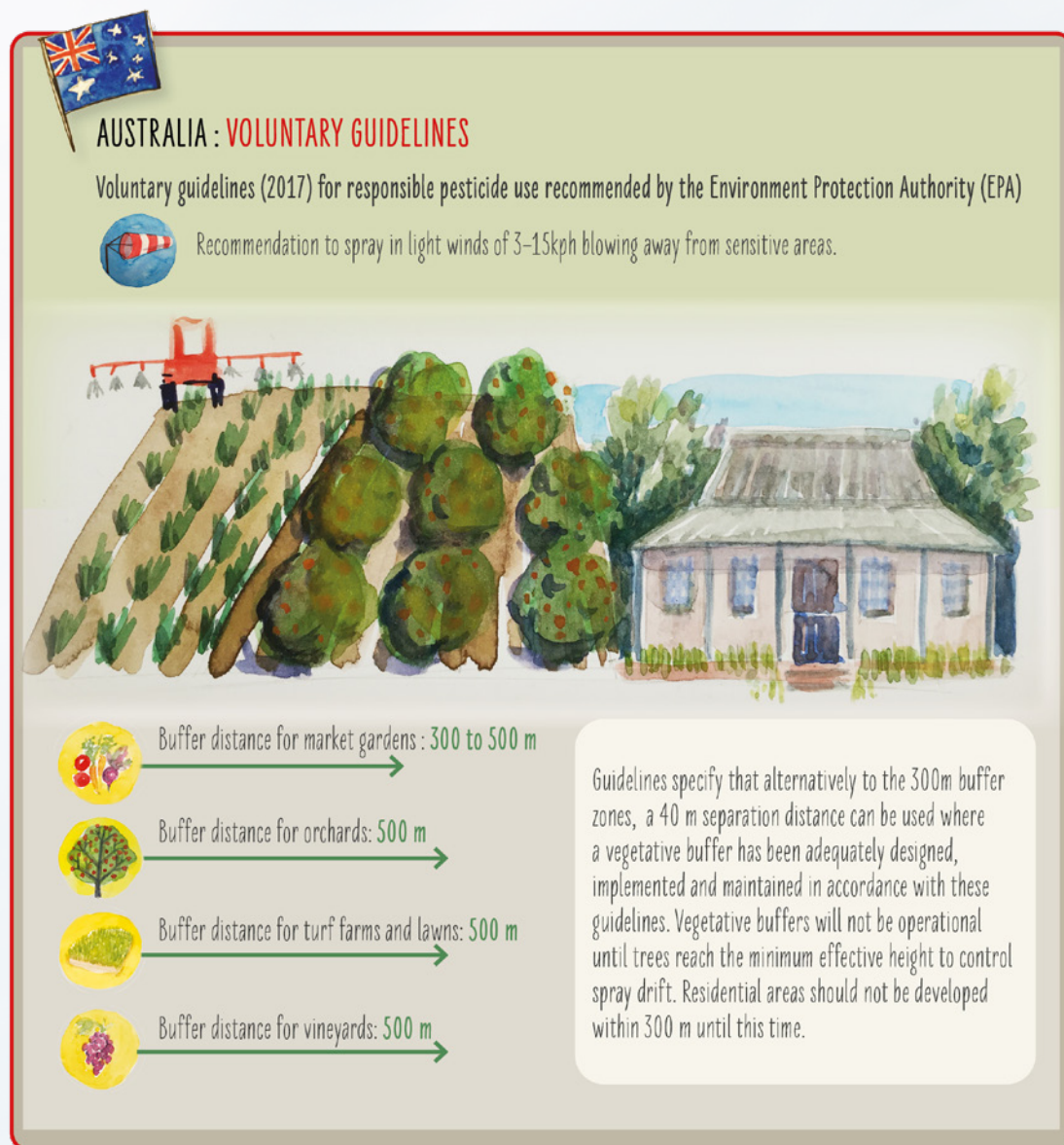


Figure 5: Australia: Voluntary guidelines for buffer zones [based on the Government of Western Australia Department of Health (n.d)]

The Government of Western Australia Department of Health published voluntary guidelines for the delineation of agricultural from residential land uses. As shown in figure 5, for any broadscale operations, the recommended buffer distance ranges from 300 to 500 metres for markets gardens, and is 500 metres for all other crops. The guidelines further specify that the precise design of the buffer depends on the chemicals used, the method of application, the site, the proposed land use, and nearby elements of land use.

8. Opportunities presented by South Africa's legislation to advocate for compulsory buffer zones

Lobbying for the 2010 Pesticide Management Act to become effective

As mentioned above, the Pesticide Management Act remains ineffective and vastly under-implemented to date. This Act is flawed in design as it assumed the need for pesticides for agricultural production and focuses on their management, as opposed to the management of pests; therefore, no room is made for non-chemical alternatives.

It is however worth pointing out aspects of the policy which practitioners should bear in mind when steps are taken to hold the legislator accountable for the provisions made in the Act. A few sections of Chapter 4 (i) are discussed hereunder:

1. "special attention should be given to pesticides that pose unmanageable risk, with an understanding that such pesticides should be considered for phasing out, severe restriction and bans. Those that will be considered include those with endocrine-disrupting properties (EDPs), Persistent Organic Pollutants (POPs), carcinogenic and immuno-toxic potential, formulations classified by WHO as Extremely Hazardous (class 1a) and Highly Hazardous (class 1b), as well as pesticides associated with frequent and severe poisoning incidents."
 Beyond the scope this article gives for phasing out dangerous pesticides, it also makes it possible to restrict or ban pesticides in specific areas. This clause could be brought forward in the context of mandatory buffer zones around sensitive areas (schools, health care centers, etc.).
2. "Considering how prevalent the use of pesticide is in ecosystems where endangered species are at risk, the DALRRD shall give explicit protection for such species."
 In certain environments where endangered species are at risk, this implies that a moratorium on spraying could be placed.
3. "Where necessary, the DAFF, through the regulatory system, shall place requirements or restrictions on users to limit the movement of pesticides to water. These will include instituting buffer zones, restricting aerial spraying in a certain proximity to water sources."
 This is an explicit mention of buffer zones, which are only called for when there is spraying close to water sources. This is limited scope yet still an avenue to create a precedent for defining buffer zones. There is scope to leverage the National Water Act for this (see below).
4. "The Policy recognizes that effective regulations are based on good science, and that strong ties are needed between research/monitoring and regulation. Research and monitoring will help to identify changes in pesticide pollution, danger spots, and particularly problematic pesticides."
 Under this provision, people should be able to argue for "good science" to be supported as an evidence base for monitoring.

Although these points are articulated as principles, without any supporting details as to how they should play out, there is scope for them to form the basis of how DALRRD could be encouraged to take agrochemical reform seriously.

Leveraging the National Water Act

The Department of Water & Sanitation (DWS) is the custodian of South Africa's water resources and therefore assumes public trusteeship of water resources, which includes watercourses, surface water, estuaries, or aquifers. The National Water Act (NWA) (Act No. 36 of 1998) aims to protect water resources, through (direct extract):

- The maintenance of the quality of the water resource to the extent that the water resources may be used in an ecologically sustainable way;
- The prevention of the degradation of the water resource; and
- The rehabilitation of the water resource.

A watercourse means:

- A river or spring;
- A natural channel in which water flows regularly or intermittently;
- A wetland, lake or dam into which, or from which, water flows; and
- Any collection of water which the Minister may, by notice in the Gazette, declare to be
- A watercourse, and a reference to a watercourse includes, where relevant, its bed and banks.

No activity may take place within a watercourse unless it is authorised by DWS. According to Section 21 (c) and (i) of the NWA, a Water Use License (WUL) is required for any activities that impede or divert the flow of water in a watercourse or alter the bed, banks, course or characteristics of a watercourse. The characteristics of a watercourse refers to the resource quality, which, according to the act is defined, *inter alia*, as: "the water quality, including the physical, chemical and biological characteristics of the water"

The regulated area of a watercourse for section 21(c) or (i) of the Act means (quoted):

- a) The outer edge of the 1 in 100-year flood line and/or delineated riparian habitat, whichever is the greatest distance, measured from the middle of the watercourse of a river, spring, natural channel, lake or dam;
- b) In the absence of a determined 1 in 100-year flood line or riparian area the area within 100 m from the edge of a watercourse where the edge of the watercourse is the first identifiable annual bank fill flood bench (subject to compliance to section 144 of the Act); or
- c) A 500 m radius from the delineated boundary (extent) of any wetland or pan.

Any water use activities that do occur within the regulated area of a watercourse must therefore be assessed using the DWS Risk Assessment Matrix (GN 509) to determine whether activities may be generally authorised (Low Risk according to the Risk Assessment Matrix) or require a WUL (Medium or High Risk according to the Risk Assessment Matrix). The premise is that any activities that have the potential to affect the Present Ecological State (PES) negatively or Ecological Importance and Sensitivity (EIS) of a watercourse would require a license. A license typically has conditions attached to mitigate and monitor the effects of the impact to the watercourse over time.

Based on the definitions described above, the application of pesticides or fertilizers to agricultural fields that lie immediately adjacent to a watercourse should qualify as a Section 21 (i) water use as the off-target movement of pesticides into adjacent water resources (e.g. via runoff and spray drift) certainly poses a potential risk of altering the characteristics of the watercourse as defined above. Based on the requirements of the Act, this implies that a risk assessment would need to be conducted to determine whether the resource quality of a watercourse is negatively affected and, if so, a WUL and associated conditions would be required. The requirement for a license provides an ideal opportunity for establishing and maintaining riparian buffers as part of the license condition.

While the definitions described above do not appear to exclude nonpoint source agricultural pollution, the current interpretation of Section 21 (i) water uses are however generally restricted to physical structures that are built within or in close proximity to watercourses. To the best of our knowledge the incorporation of nonpoint agricultural pollution as a legitimate Section 21 (i) water use has not yet been tested. Assuming that the interpretation above is correct, it would mean that all farms with fields adjacent to a watercourse would require a license for a Section 21 (i) water use. The majority of farms in South Africa would however most likely qualify as an Existing Lawful Use (ELU) which means that the use of water would be subject to any applicable legislation for a period of two years before the commencement of the NWA (i.e. before October 1998). However, newly established farms or even farms that change from one crop type to another (which would require the application of a different suite of pesticides) should, based on the definitions described above, theoretically require a WUL for the application of pesticides. The Water Act could thus be leveraged as an entry point to legislate buffer zones in the country.

Leveraging the Air Quality Act

There is no mention of pesticides in the 2004 Air Quality Act.²² However Section 15 (2) of the Act requires district municipal authorities to include local “air quality management plans” in their integrated development plans (IDPs), as per chapter 5 of the Municipal Systems Act.

This represents an opportunity for local communities to hold the municipality accountable for air quality planning and monitoring. Local communities can engage with their district municipality and demand to be recognised as stakeholders in the determination of these local air quality management plans.

Specific elements that could be referenced with respect to pesticide drift include the following objectives of air quality management plans (Section 16 of the Act):

- (iii) to identify and reduce the negative impact on human health and the environment of poor air quality;
- (vi) to address the effects of emissions from any point or non-point source of air pollution

It is in the context of such a plan that district-level negotiations should be held, to demand mandatory buffer zones for specific areas, until such time as mandatory buffer zones are imposed nationally. Ideally these requests should be informed by community monitoring of spray drift in the area concerned. Pesticide drift catchers can be used to monitor pesticides in the air and to assess health risks.

22 See: https://www.environment.gov.za/sites/default/files/legislations/nema_amendment_act39.pdf

Conclusion

A South African campaign against pesticide spray drift should be requesting implementation decrees that would see aspects of the 2010 Pesticide Management Policy implemented.

The specific demands made should include:

1. taking proactive measures **to minimize or prohibit pesticide use** in specific areas (residential areas, schools, healthcare centres, business centres, public spaces, water bodies)
2. **mandating compulsory buffer areas** to serve as buffer zones. Their dimensions should be determined by methodologies relevant to and feasible in the South African context. The National Water Act constitutes a potential leverage point to create a precedent with buffer zones that safeguard zones for surface and ground-water used for the abstraction of drinking water. Buffer zones should be incorporated in integrated development plans at the level of the municipality as a means to prevent and manage conflicting land uses, as was amply illustrated by the case studies
3. **banning aerial spraying of pesticides**
4. implementing stricter policies and procedures with regards to **chemical handling to prevent spillage**
5. **implementing neighbour notification laws for all pesticide applications** in residential areas, schools, healthcare centres, business centres, public spaces, and requesting from farmers **compulsory access to information** on what is being sprayed
6. **supporting**, through a dedicated fund, **farmers transitioning away from chemical pesticides**
7. identifying means to reduce the risk of spray drift by aiding farmers in the identification of **spray-reducing equipment**
8. **penalising abuse and setting compensation mechanisms** for spray drift victims so that social and environmental justice can prevail

These aspects should form an integral part of a structural and conceptual reform of the legal framework regulating agrochemicals in South Africa.

Annexures

Annexure 1. Brief review of a legislative agrochemical framework in South Africa and what affected communities can do in the case of spray drift

Until such time the South African Government bans hazardous pesticides, reins in the abusive spraying of pesticides and mandates buffer zones, affected communities will have to make use of the limited legal framework regulating pesticide to have their right to a clean environment protected. The following is an adapted version of the recommendations provided by the TATIB Foundation.²³

Jurgen Schirmacher reported making successful use of two laws to win several High Court cases against a farmer that was using agricultural chemicals in an irresponsible and unlawful manner.

FARM FEEDS, FERTILISERS AND STOCK REMEDIES ACT 36 OF 1947

Each and every agricultural chemical (pesticide / fungicide/ herbicide / fertilizer) that is used, and sold in South Africa has to be registered under Act 36 of 1947. One of the main clauses within Act 36 that he used in his successful High Court cases, pertained to a provision in the Act that states that when someone uses an agricultural chemical, the person has to follow the product label instructions and special warnings scrupulously. The product label is often attached to the product and is normally a few pages long. It can also be downloaded from the internet. The product label will have special warnings and instructions that generally are worded as follow:

“Avoid spray drift of the product into/onto areas that are not under treatment”

“Avoid run off into adjacent waterways, rivers and dams”

“Notify all the inhabitants adjacent to the area being sprayed and issue the necessary warnings”

“Do not spray in windy conditions”

“Do not enter the area that has been sprayed for x number of hours”

PESTICIDE MANAGEMENT ACT (2010/2011)

Within this act there is mention of having to erect prescribed warning signs, prior, during and after spraying, so as to warn people that a certain area has been sprayed so that they know to stay out of the area. This is useful when for example a municipality sprays parks, verges, pavements, etc. with a herbicide. By law they (the municipality) need to put signs up so that people don't go and sit on or come into contact with freshly sprayed areas. There is also a clause within this act that anyone who applies an agricultural chemical in a commercial way (farmers, pest control companies, municipalities, golf courses etc.) needs to be a registered pest control officer (PCO).

THE CONSTITUTION OF THE REPUBLIC OF SOUTH AFRICA

The Constitution states that each and every citizen of this country has the right to live in an environment that is not harmful to his and her health and well-being. So if someone is caught in spray drift, his/her rights under the constitution are protected.

How to establish whether the pesticide is lawful or unlawful and what to do should you be affected by spray drift (extract):²⁴

1. Can you smell the product being used? If yes, then spray drift in the form of fine particles/droplets is in the air and Act 36 of 1947 has been broken.
2. Have any signs been put up to warn the public?

23 Edited from: <https://tatibfoundation.blogspot.com>

24 Also available at: <https://gmopoisonfreezones.org.za/action/how-to-stop-spraying-in-your-area/>

3. Who is doing the spraying? A company? An employee? A farmer? A contractor for the municipality?
4. Are they wearing protective clothing e.g. overalls, gloves, masks, goggles? If not, then they are in contravention of Act 36 and various Occupational Health and Safety Acts.
5. Is it windy? If yes, then there might be spray drift out of the area being treated.
6. Is there imminent rain? If so then the rain will wash the product into storm water drains and contaminate rivers, streams and dams.

GUIDELINES TO STOP SPRAY DRIFT OF HAZARDOUS AGROCHEMICALS IN YOUR NEIGHBOURHOOD:

7. If it is safe to do so, take photos/ videos of the spraying – look for signs of spray drift to show that the product has left the areas being treated. Look for innocent bystanders (e.g. mothers, children, the elderly) in the area who might be exposed to the product. Note the time and date.
8. Try to take or obtain photos of the spraying method, product containers, the company or people doing the spraying, the area being sprayed.
9. If safe to do so, approach the sprayers and ask them WHY they are spraying, WHAT product they are using, and WHO told them to do the spraying. Try to get names.
10. Contact the head office of the company/municipality (usually under Parks & Cemeteries) doing the spraying by phone/email to find out the name of the director/manager.
11. Voice your concerns about the spraying as this is your Constitutional right as a citizen. If the workers doing the spraying refuse to tell you, then ask the manager what was being sprayed. The person may refuse, which could send out a message that they have something to hide. You can also inform them that if they refuse to tell you, then you will serve them with a PAIA which would force them, by law, to give you the information to which you have a right. After making the phone call, draft a letter to this person, outlining in point form what you witnessed. State what you want – i.e. the name of the product being used, and under whose authority. If it was the local city council, then the name of the official who placed the order. You can also ask if they are a registered pesticide officer/applicator, as required by law. By law, they must furnish you with their registration details.
12. Then phone the company/person who placed the order (e.g. municipality). Repeat step 10 & 11. Put everything in writing in order to establish a paper trail.
13. They will tell you that: a) the product is registered for use in South Africa b) the product is safe for use and poses no threat to human and environmental health, IF USED CORRECTLY. You are trying to establish if the product IS being used correctly. IF NOT, then it poses a threat to human and environmental health. You could also ask that they furnish you with the material safety data sheets (MSDS) and the product label document.
14. Everyone, including municipalities, government bodies, schools etc. has a duty to ensure that you are not exposed to anything that may be detrimental to your health and well-being when you are on their premises. That is your right!
15. If the municipality is doing the spraying, then copy the municipal manager, the relevant health official and public works manager in via email. Phone and email the district municipality, repeating the above steps. Put everything in writing and document all phone calls, then email those details to them. Copy in your local councillor. This prevents public servants from passing the buck.
16. Report the case to your LOCAL POLICE STATION. Your affidavit must state what happened and how you were affected. Attach a medical report if you managed to get one. This is the only way to have a DAFF officer come and investigate the case.
17. Be polite but firm. ALWAYS send an email after your conversations, confirming what you spoke about. Do your homework. Be informed. Know your rights. Know the product to show that it has been used in contravention of the law. They have the right to spray/use a registered product, BUT they must follow the law scrupulously so that they do not put anyone at risk. Look for ways where the correct procedures have not

been followed. You don't have to prove that the product is not safe. All you have to do is show that laws have been broken.

GUIDELINES FOR PARENTS, TEACHERS, CONCERNED CITIZENS TO STOP THE SPRAYING OF POISONOUS AGROCHEMICALS IN YOUR NEIGHBOURHOOD, SCHOOL, COMMUNITY:

18. Even if the spraying is conducted lawfully in your neighbourhood or child's school, and you want it to stop, write a letter to the neighbours or parents. Ask them to sign an agreement that alternatives to the poisons must be made available (cutting, weeding by hand, ground covers etc.). It's important that your community is on board with you.
19. Attend community meetings. Speak to your ward councillor. Follow up with an email of your chat. If spraying is at a school, address the issue at a parent meeting.
20. Remember that the municipality or school is providing a service to you for which you are paying. If you know of a trusted garden services that can do the job without poisons, refer them as a back-up. They may not use these services, but the present contractors may be asked to change their methods.
21. Send reminders, stating the time by which you expect your reply. They might ask you to go in and present the problem. Use your correspondence, product label, WHO classification, photos/videos etc. Don't talk about anything that you are not sure of. Remember that you don't need to be a scientist or lawyer to have a valid concern. As a parent, teacher, caregiver or child, it is your right to feel safe in your environment. Even if there is a slight concern, cough or allergy, precaution should be taken.

Annexure 2. Equations to determine exposure levels using the AGricultural DISPersal model

Equation 1: Exposure of infants after application to turf

$$\text{Exposure} = \frac{(\text{Dermal adsorption}) + (\text{Oral adsorption})}{\text{Body Weight}}$$

$$\text{Exposure} = \frac{(AR \times DepR \times Ac \times DA \times TC \times ET) + (AR \times DepR \times Ac \times B \times SAo \times FQ \times ET_o)}{BW}$$

$$\text{Acceptable Risk If: Exposure} \leq \frac{NOAEL}{MoE}$$

$$\text{Unacceptable Risk If: Exposure} > \frac{NOAEL}{MoE}$$

For applications to turf, the application rate (AR) is the level of active ingredient deposited in the target area. When considering exposure of an off-target area through spray drift (which varies according to the way the product is applied as well as the application rate), this value is equivalent to the level of active ingredient that can be deposited outside the target area and not pose an unacceptable risk (i.e. the bystander RAL).

As 'Exposure' is equal to, or less than, the No Observed Adverse Effect Level (NOAEL) divided by the Margin of Exposure (MoE), the following substitutions can be made to this formula which can then be converted to calculate the RAL, as shown in Equation 2:

- AR = RAL, which becomes the unknown in the formula
- Exposure = NOAEL/MoE, which become known variables in the formula

Equation 2: Conversion of application to turf formula to determine bystander area RAL

$$\frac{NOAEL}{MoE} = \frac{(RAL \times DepR \times Ac \times DA \times TC \times ET) + (RAL \times DepR \times Ac \times B \times SAo \times FQ \times ETo)}{BW}$$

$$\frac{NOAEL}{MoE} \times BW = (RAL \times DepR \times Ac \times DA \times TC \times ET) + (RAL \times DepR \times Ac \times B \times SAo \times FQ \times ETo)$$

$$\frac{NOAEL}{MoE} \times BW = RAL \times ((DepR \times Ac \times DA \times TC \times ET) + (DepR \times Ac \times B \times SAo \times FQ \times ETo))$$

$$RAL = \frac{\left(\frac{NOAEL}{MoE}\right) \times BW}{((DepR \times Ac \times DA \times TC \times ET) + (DepR \times Ac \times B \times SAo \times FQ \times ETo))}$$

The variables in this formula and any default values are explained in the table below.

Variable		Unit	Default value	Comment
AR	Application rate	mg ac/cm ²	1.0–DA	Deposition on turf calculated from the application rate for the use pattern on the label.
B	Oral bioavailability	fraction	1.0–DA	Estimates of oral bioavailability are usually determined from PK studies with oral dosing. In the absence of pesticide-specific data, suitable surrogate data using a valid scientific argument, such as 'read across' would be acceptable.
BW	Bodyweight	Kg	11 (1–2 year olds) OR 15 (2–3 year olds)	EnHealth (2012) recommends a bodyweight value for 1-2 -year-old children of 11 kg, and for 2–3 year olds of 15 kg.
DA	Dermal absorption factor	Fraction	1.0	In the absence of product-specific data, suitable surrogate data using a valid scientific argument, such as 'read across' can be used.

Variable		Unit	Default value	Comment
DepR	Deposited Residue	Fraction	Combined as 0.05	Together, these factors constitute a transferable residue factor (DepR x Ac). EnHealth (2012) states that default for the accessibility of organics on surfaces is 100% (1.0), and 0.01% (0.0001) for inorganics, and that the default values are to be used 'only when other reasonable information is not available'. On the basis that most agricultural chemicals are organics, the default Ac is 1.0. The default value for the DepR is 0.05, which is consistent with upper limits of US EPA (1997) so is considered to be derived from 'reasonable information'. Therefore, without additional information, a default transferable residue factor of 0.05 for turf is used.
Ac	Accessibility factor	Fraction		
ET	Exposure time per day	hours per day	1.1 (1–2 year olds) OR 1 (2–3 year olds)	For time spent playing on turf, EnHealth (2012) recommends using the mean values of 1.1 [95th percentile of 2 hours] and 1 hour [95th percentile of 2 hours] for infants (1-<2) and toddlers (2-<3) respectively.
ETo	Duration of hand-to-mouth actions per day	Hours per day	1.1 (1–2 year olds) OR 1 (2–3 year olds)	EnHealth (2012) indicates that the duration of mouthing for children is highly variable depending on the age group and the type of objects included in the estimates. Therefore, in the absence of specific mouthing duration of the hands (including fingers), a default value equivalent to the ET (Exposure time per day) is used.
FQ	Hand-to-mouth action frequency	Contact events per hour	14 (1–2 year olds) OR 5 (2–3 year olds)	EnHealth (2012) state that the default mouthing frequency values in outdoor settings are 14 or 5 contacts per hour for 1–2 or 2–3 year olds, respectively.
MoE	Margin of Exposure	factor	100	The MoE is the ratio of no-observed-adverse-effect-level (NOAEL) for the critical effect to the estimated exposure dose or concentration of a substance, as stated in OECD (2003). A MoE of 100 or higher is considered acceptable for this risk assessment. This comprises of a 10-fold safety factor for both potential intraspecies variability and interspecies variability (ie $10 \times 10 = 100$).

Variable		Unit	Default value	Comment
NOAEL	No Observed Adverse Effect Level	mg/kg	-	No-observed-adverse-effect level (NOAEL): greatest concentration or amount of a substance, found by experiment or observation, that causes no detectable adverse alteration of morphology, functional capacity, growth, development, or lifespan of the target organism under defined conditions of exposure. WHO (2009) Environmental Health Criteria, No. 240.
SAo	Surface area potentially exposed from hand-to-mouth actions	cm ² per contact event	19	The value of 19 cm ² per contact event is used for area mouthed per mouthing event. This is based on US EPA (1997) where a value of 0.127 for the fraction of hand surface area mouthed is used and in combination with EnHealth (2012) value for the surface area of a single hand (150 cm ²) (150 cm x 0.127), the area of a hand (including fingers) mouthed equals 19.55 cm ² , rounded to 19 cm ² .
TC	Transfer coefficient	cm ² /hr	7,073 (1–2 year olds) OR 8,700 (2–3 year olds)	The default TC value for 2–3 year olds playing on turf is 8700 cm ² /hr which is the US EPA (1997) upper percentile value for toddlers playing on turf. Extrapolating the value to 1–2 year olds based on a 1.23–fold lower exposed surface area vs 2–3 year olds gives a TC of 7073 cm ² /hr.

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